

Book XI of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Chapter 4

Proposition XI.4

4.1 Statement

Euclid XI.4 gives the basic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The formal reconstruction is implemented in

```
CGJteamLab/Proposition11_4.lean
```

on top of the spatial infrastructure in

```
CGJteamLab/Hilbert3DAxioms.lean  
CGJteamLab/Hilbert3DInterface.lean
```

The Lean source is authoritative for theorem signatures, helper names, actual dependencies, and proof status.

4.2 Euclid XI.3 and the target notion

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt  
  [H : HilbertIncidence Geo]  
  (l m : Geo.Line)  
  (O : Geo.Point) : Prop :=  
  H.OnLine O l /\  
  H.OnLine O m /\  
  exists A B : Geo.Point,  
    Ne A O /\  
    Ne B O /\
```

```

H.OnLine A l /\
H.OnLine B m /\
Not (PrimCollinear Geo A O B) /\
HilbertRightAngle Geo A O B

```

The explicit noncollinearity condition is important. It prevents a degenerate “self-perpendicular” witness and yields the reusable theorem

```

hilbert_linesPerpendicularAt_ne

```

showing that perpendicular lines are distinct.

For a line and a plane:

```

def HilbertLinePerpendicularPlaneAt
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point) : Prop :=
H.OnLine O l /\
S.OnPlane O pi /\
forall m : Geo.Line,
  HilbertLineInPlane Geo m pi ->
  H.OnLine O m ->
  HilbertLinesPerpendicularAt Geo l m O

```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

4.3 Classical configuration

Let m, n be two distinct straight lines of a plane π , meeting at O , and let l be a line through O perpendicular to both.

The desired conclusion is that l is perpendicular to every line $q \subset \pi$ through O .

4.4 Euclid’s classical proof

Euclid makes the argument constructive.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where E is the common point of the two given lines. Draw an arbitrary straight line GEH through E in the plane, choose an arbitrary point F on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

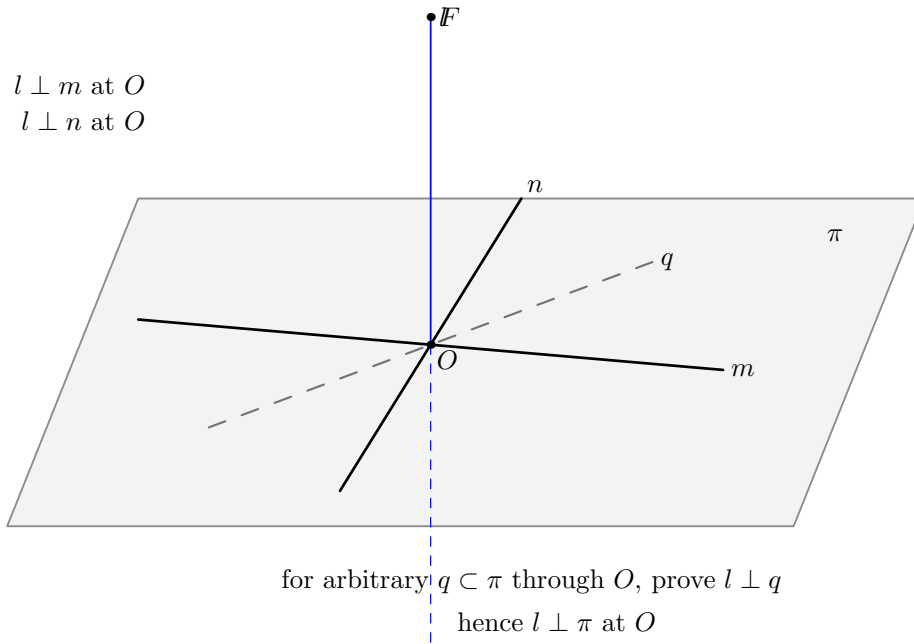


Figure 4.1: Classical configuration for Euclid XI.4. The ambient line l passes through O and is perpendicular at O to two distinct lines m, n lying in the plane π . The dashed line q represents an arbitrary line of π through O , which is the universal target of the conclusion.

1. Vertical angles at E are equal (I.15). Together with the four equal radial segments, I.4 gives

$$AD = CB$$

and a corresponding angle equality.

2. Applying I.26 to the planar transversal gives

$$GE = EH, \quad AG = BH.$$

3. Since FE is perpendicular to both original plane lines, I.4 applied to two right triangles gives

$$FA = FB, \quad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.

5. A further I.4 gives

$$FG = FH.$$

6. Finally, from

$$GE = EH, \quad FE = FE, \quad FG = FH,$$

I.8 yields

$$\angle GEF = \angle HEF.$$

Since $G - E - H$, the adjacent equal angles are right angles.

7. The transversal GEH was arbitrary. Hence FE makes a right angle with every line through E in the plane. By XI.Def.3, $FE \perp \pi$.

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

4.5 Formal theorem

The production theorem is:

```

theorem euclid_proposition_11_4
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (m n : PlaneLine Geo pi)
  (l : Geo.Line)
  (O : PlanePoint Geo pi)
  (hmn : Ne m n)
  (hPerpM :
    HilbertLinesPerpendicularAt Geo l m.1 O.1)
  (hPerpN :
    HilbertLinesPerpendicularAt Geo l n.1 O.1) :
  HilbertLinePerpendicularPlaneAt Geo l pi O.1

```

The theorem does not assume separately that $l \neq m$ or $l \neq n$. Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

4.6 Spatial architecture

4.6.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\text{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```

PlanePoint Geo pi
PlaneLine   Geo pi

```

and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

4.6.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid’s unqualified use of plane triangle congruence inside solid geometry.

4.7 The balanced cross

The first construction is the formal analogue of Euclid’s

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

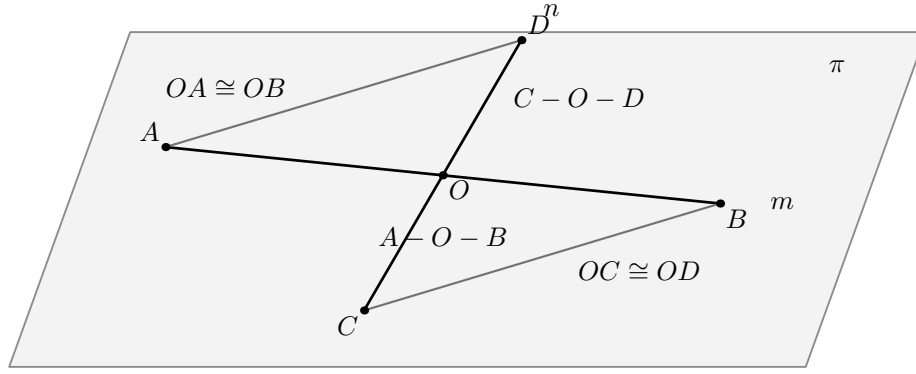
```
hilbert_XI4_balanced_cross_exists
```

and points A, B, C, D satisfying

$$A - O - B, \quad C - O - D$$

and

$$OA \cong OR, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$



all objects in this figure lie in π

Figure 4.2: Balanced cross in π . Points A, B lie on m , points C, D lie on n , with $A - O - B$, $C - O - D$, $OA \cong OB$, and $OC \cong OD$. The connector segments AD and BC are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \quad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

4.8 Fixed-transversal prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \quad B - J - C, \quad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \quad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point $F \in l$, together with the two original perpendicularities, then gives

$$AF \cong BF, \quad CF \cong DF.$$

The proof packages this in

```

hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant

```

The spatial SSS/SAS chain is then:

```

hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal

```

Its mathematical content is

$$\begin{aligned}
& AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\
& \quad \Downarrow \\
& \angle DAF \cong \angle CBF \\
& \quad \Downarrow \\
& A - G - D, \quad B - J - C \\
& \quad \Downarrow \\
& \angle GAF \cong \angle JBF \\
& \quad \Downarrow \\
& AG \cong BJ, \quad AF \cong BF \\
& \quad \Downarrow \\
& GF \cong JF \\
& \quad \Downarrow \\
& OG \cong OJ, \quad OF \cong OF, \quad GF \cong JF \\
& \quad \Downarrow \\
& \angle GOF \cong \angle JOF \\
& \quad \Downarrow \\
& G - O - J \\
& \quad \Downarrow \\
& \text{RightAngle}(G, O, F).
\end{aligned}$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

4.9 The arbitrary transversal: the real formal difficulty

The public conclusion quantifies over *every* line of π through O . A conventional drawing suggests that such a transversal meets the two connector segments AD and BC , but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \quad \text{or} \quad BD/AC.$$

For a transversal $q \neq m, n$, apply Pasch to the triangle ABD . The line q enters through AB at O , hence it exits either through AD or through BD .

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point J opposite G on q with

$$G - O - J, \quad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```
hilbert_XI4_first_exit_opposite_lands_on_BC
hilbert_XI4_second_exit_opposite_lands_on_AC
```

so that the opposite point lands on the correct paired connector. The combined package is

```
hilbert_XI4_arbitrary_transversal_connector_package
```

and returns

$$\begin{aligned} G, J \in q, \quad G - O - J, \quad OG \cong OJ, \\ \text{and either} \\ A - G - D, \quad B - J - C, \\ \text{or} \\ B - G - D, \quad A - J - C. \end{aligned}$$

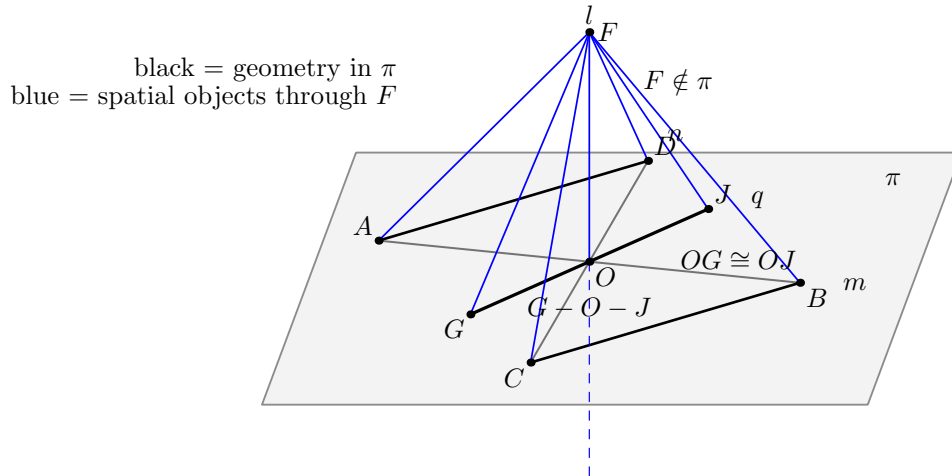


Figure 4.3: Arbitrary-transversal stage. For $q \subset \pi$ through O , Pasch selects one of the two connector families. The opposite point J is constructed with $G - O - J$ and $OG \cong OJ$. The spatial SSS/SAS chain then proves that $\angle GOF$ is right.

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

4.10 Right angle on every transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```
hilbert_XI4_arbitrary_transversal_right_angle
```

states, in substance:

$$\begin{aligned} q \subset \pi, \quad O \in q, \quad q \neq m, n, \\ F \notin \pi, \quad AF \cong BF, \quad CF \cong DF \end{aligned}$$

imply the existence of $G \in q$, $G \neq O$, such that

$$\text{RightAngle}(G, O, F).$$

The two exceptional cases $q = m$ and $q = n$ are discharged directly from the hypotheses of XI.4.

4.11 Why the point F is outside the plane

The core proof chooses an arbitrary nonvertex point F on l . To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume $F \in \pi$. Then the line l , determined by O and F , is itself a line of π . Transfer the two ambient perpendicularities to **PlaneGeo** **pi**. Choose nonvertex points $A \in m$ and $C \in n$. Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that A, O, C are collinear. Line uniqueness then forces $m = n$, contradicting the hypothesis.

No parallel postulate is used.

4.12 Closing the universal line–plane statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose $F \neq O$ on l ;
2. construct the balanced cross A, B, C, D in π ;
3. prove $F \notin \pi$;
4. prove $AF \cong BF$ and $CF \cong DF$;
5. let q be an arbitrary line of π through O ;
6. if $q = m$ or $q = n$, use the hypotheses;
7. otherwise invoke the arbitrary-transversal theorem and obtain a point $G \in q$ such that $\angle GOF$ is right;
8. construct the witness required by `HilbertLinesPerpendicularAt Geo 1 q 0`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since l and q intersect at O , they determine an explicit plane ρ . Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

4.13 Complete formal DAG

The proof architecture can be summarized as follows.


```

two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD = BC and angle OAD = angle OBC
-> choose F != O on l
-> F notin pi
-> AF = BF, CF = DF
-> arbitrary q in pi through O
-> if q=m or q=n: hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG = OJ
-> landing on BC or AC
-> ASA: AG = BJ (or BG = AJ)
-> spatial SSS/SAS: GF = JF
-> spatial SSS: angle GOF = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.

```

4.14 Classical dependencies versus Lean dependencies

Role	Euclid	Lean reconstruction
Four equal radial segments	cut off AE, EB, CE, ED equal	opposite-point construction on each of the two lines inside <code>PlaneGeo pi</code>
Vertical angles	I.15	developed vertical-angle and angle-congruence infrastructure
First triangle congruence	I.4	balanced-cross SAS helper
Transversal equalities	I.26	Pasch, transversal ASA, and the two landing branches
Distances from F	I.4 on right triangles	spatial perpendicular-bisector/equidistance helper
Cross-plane triangle comparison	I.8 and I.4	general spatial SSS/SAS API through explicit carrier planes
Arbitrary line in the plane	“drawn at random”	universal q ; cases $q = m$, $q = n$, or the Pasch branch
Line perpendicular to plane	XI.Def.3	the universal line-plane perpendicularity predicate

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

4.15 Axiomatic status

The public theorem assumes the spatial Hilbert hierarchy

```

HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo

```

```
HilbertSpaceOrder
HilbertSpaceCongruence
```

No `HilbertEuclideanPlane` instance occurs in the theorem signature. In particular, the proof of XI.4 does not use Hilbert Group IV, the Euclidean parallel axiom.

No continuity or Archimedean axiom is introduced by XI.4.

The final Lean audit is:

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[proptext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this stage. XI.4 introduces no new proposition-specific axiom.

4.16 Formalization notes

4.16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

4.16.2 The point F is not a normal-vector surrogate

The point F is merely a nonvertex point chosen on the candidate line l . It is used to form ordinary synthetic triangles such as AOF , BOF , GOF , and JOF . The proof never assigns coordinates to F and never interprets OF as a vector.

4.16.3 Why PlaneGeo is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \longleftrightarrow \text{PlaneGeo}(\pi) \text{ or } \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

4.16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$ and $C - O - D$ are strict orders;
- the auxiliary point F is outside π ;
- a general transversal need not meet the fixed pair AD, BC ;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through l and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

4.16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through O meets both segments AD and BC . The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle ABD . This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \text{ or } (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

4.17 Reusable infrastructure produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

4.18 Dependency summary

Hilbert spatial incidence + order + congruence
 $\Downarrow$
 PlaneGeo slices and ambient/plane bridges
 $\Downarrow$
 nondegenerate line-line perpendicularity
 $\Downarrow$
 balanced cross $A - O - B, C - O - D$
 $\Downarrow$
 $AD \cong BC, \angle OAD \cong \angle OBC$
 $\Downarrow$
 $F \in l, F \neq O, F \notin \pi$
 $\Downarrow$
 $AF \cong BF, CF \cong DF$
 $\Downarrow$
 Pasch for arbitrary $q \subset \pi$ through O
 $\Downarrow$
 two connector branches + opposite point J
 $\Downarrow$
 $GF \cong JF, \angle GOF \cong \angle JOF$
 $\Downarrow$
 $\text{RightAngle}(G, O, F)$
 $\Downarrow$
 $l \perp q$ for every $q \subset \pi$ through O
 $\Downarrow$
`euclid_proposition_11_4.`

4.19 Status

Production theorem	<code>Geometry.euclid_proposition_11_4</code>
Production file	<code>CGJteamLab/Proposition11_4.lean</code>
General 3D interface	<code>CGJteamLab/Hilbert3DInterface.lean</code>
Coordinates/vectors	no
Euclidean parallel axiom (Group IV)	no
Continuity	no new use
New XI.4-specific axiom	no
Full project build	clean
Axiom audit	completed

4.20 Conclusion

Euclid XI.4 is the first major test of the project’s spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid’s proof: equal-segment constructions, vertical angles, triangle congruence, an arbitrary transversal, and the definition of a perpendicular to a plane.

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate line is perpendicular to every line of the given plane through the common point. In that sense the proof is not a modern replacement of XI.4 but a detailed synthetic reconstruction of its logical content.

Chapter 5

Proposition XI.5

5.1 Introduction

Proposition XI.5 is the first Book XI proposition whose conclusion is a genuinely spatial coplanarity statement forced by perpendicularity.

A line l passes through a point O and is perpendicular there to three lines m, n, p . Two of these lines, m and n , are assumed distinct. The conclusion is that m, n, p lie in one plane.

The formal reconstruction is not a line-by-line transcription of Euclid's proof. The current spatial interface makes it possible to use Proposition XI.4 twice and thereby obtain a short contradiction argument.

5.2 Euclid's Statement

Euclid's Proposition XI.5 states:

If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

In Euclid's notation the common perpendicular is AB , and the three lines through the common point B are BC , BD , and BE .

5.3 Classical Configuration

Euclid assumes, for contradiction, that two of the three lines, say BD and BE , lie in a reference plane while BC lies in a more elevated plane.

He then considers the plane through AB and BC . Its intersection with the reference plane is a line BF . Thus AB, BC, BF lie in one plane.

At the same time, since AB is perpendicular to BD and BE , Proposition XI.4 implies that AB is perpendicular to the reference plane. Therefore AB is also perpendicular to BF .

The contradiction is that, in the plane through AB and BC , the lines BC and BF would both form a right angle with AB at the same point.

5.4 Classical Proof

The local classical dependency pattern is:

$$\text{XI.3} \longrightarrow \text{intersection line } BF,$$

$$\text{XI.4} \longrightarrow AB \perp (\text{reference plane}),$$

followed by Euclid's Definition XI.3, which turns line-plane perpendicularity into perpendicularity to every line of the plane meeting the perpendicular.

Thus the classical proof is a contradiction proof based on the intersection of two planes and on the impossibility of two distinct perpendiculars to the same line at the same point in one plane.

5.5 What is genuinely three-dimensional here?

The essential spatial content is not the elementary fact about right angles. It is the management of planes.

The proof must establish:

- a plane through two intersecting lines;
- a second plane through another pair of intersecting lines;
- that these two planes are distinct;
- a line of intersection of the two planes;
- containment of relevant lines in the appropriate planes.

The current Lean proof does not open a new planar `PlaneGeo` argument of its own. Instead, its planar geometric content is inherited through the already proved Proposition XI.4.

5.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_5
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
```

```

[_HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
(1 m n p : Geo.Line)
(0 : Geo.Point)
(hmn : Ne m n)
(hPerpM :
  HilbertLinesPerpendicularAt Geo 1 m 0)
(hPerpN :
  HilbertLinesPerpendicularAt Geo 1 n 0)
(hPerpP :
  HilbertLinesPerpendicularAt Geo 1 p 0) :
exists pi : S.Plane,
  HilbertLineInPlane Geo m pi /\
  HilbertLineInPlane Geo n pi /\
  HilbertLineInPlane Geo p pi

```

The conclusion is constructive at the level of incidence: Lean returns an explicit plane π containing all three lines.

The hypothesis `hmn : Ne m n` makes explicit a point that is usually left implicit in the classical diagram. The two intersecting lines m, n must be distinct in order to determine the reference plane.

5.7 Spatial / Planar Architecture Used

The proposition uses the ambient three-dimensional layer:

- `HilbertSpacePrimitive` for planes and point–plane incidence;
- `HilbertSpaceIncidence` for spatial plane constructions;
- `HilbertLineInPlane` for line–plane containment;
- `HilbertLinesPerpendicularAt` for spatial line–line perpendicularity;
- `HilbertLinePerpendicularPlaneAt` for line–plane perpendicularity.

The ambient geometry does not install the planar order and congruence classes as global instances on the spatial object. The architectural distinction is:

```

-- not global ambient instances:
HilbertOrder Geo
HilbertCongruence Geo

-- spatial layers:
HilbertSpaceOrder
HilbertSpaceCongruence

```

Planar order and congruence enter only through the established `PlaneGeo` machinery used inside Proposition XI.4.

5.8 Proof Architecture of the Reconstruction

Let m and n be the first two lines perpendicular to l at O .

5.8.1 Step 1: construct the reference plane

Since $m \neq n$ and both pass through O , the theorem

```
hilbert_plane_through_two_intersecting_lines
```

constructs a plane π containing m and n .

5.8.2 Step 2: apply Proposition XI.4

The lines m and n are packaged as `PlaneLines` of π , and O as a `PlanePoint` of π .

Proposition XI.4 then yields

$$l \perp \pi \quad \text{at } O.$$

In Lean this is the call to

```
euclid_proposition_11_4
```

with the two given perpendicularities $l \perp m$ and $l \perp n$.

5.8.3 Step 3: split on whether p already lies in π

If

$$p \subset \pi,$$

the theorem is finished immediately.

Assume therefore

$$p \not\subset \pi.$$

Since $l \perp p$, the lines l and p are distinct. They meet at O , so they determine another plane σ .

5.8.4 Step 4: intersect the two planes

The planes π and σ must be distinct. If $\pi = \sigma$, then $p \subset \sigma$ would imply $p \subset \pi$, contrary to the case assumption.

The theorem

```
hilbert_plane_intersection_line
```

therefore gives a line q through O contained in both planes:

$$q \subset \pi, \quad q \subset \sigma.$$

Since $l \perp \pi$ and q is a line of π through O ,

$$l \perp q.$$

This is obtained directly from

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

5.8.5 Step 5: the second use of XI.4

Now p and q both lie in σ , and

$$l \perp p, \quad l \perp q.$$

Suppose $p \neq q$. Applying Proposition XI.4 inside σ gives

$$l \perp \sigma.$$

But $l \subset \sigma$ by construction. This is impossible.

Hence

$$p = q.$$

Since $q \subset \pi$, it follows that $p \subset \pi$, and therefore

$$m, n, p \subset \pi.$$

5.9 Lean Formalization

The small proposition-local helper is:

```
theorem hilbert_linePerpendicularPlaneAt_not_in_plane
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l : Geo.Line)
  (pi : S.Plane)
  (O : Geo.Point)
```

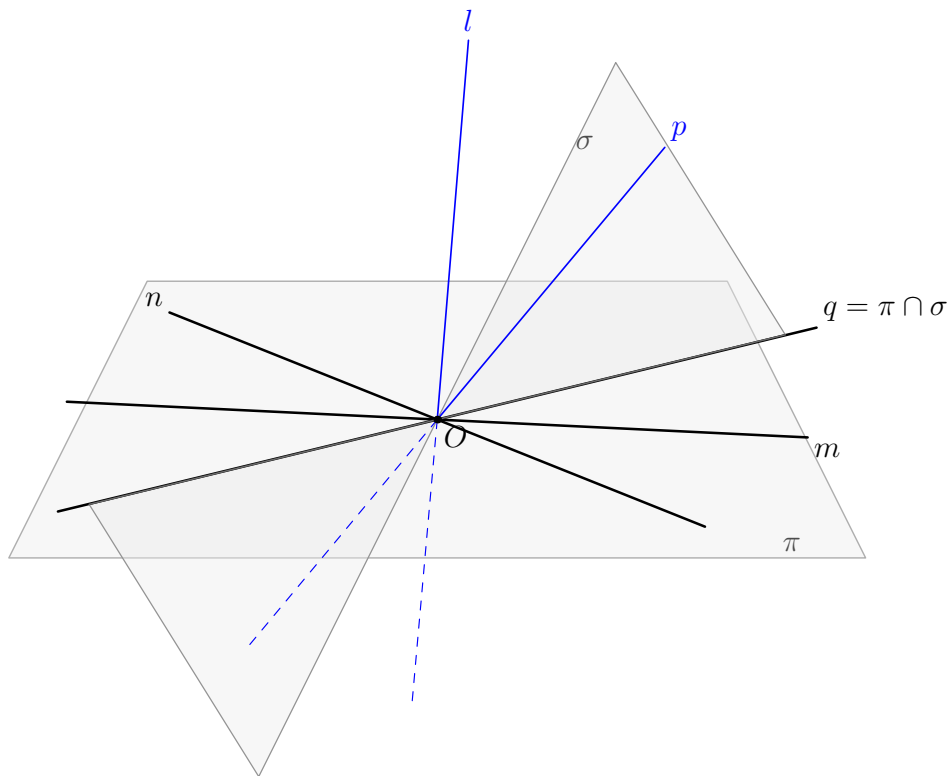


Figure 5.1: The formal contradiction argument for Proposition XI.5. The black lines m, n, q lie in the reference plane π . The spatial lines l, p are blue. Under the temporary assumption $p \not\subset \pi$, the lines l, p determine the second plane σ , and $q = \pi \cap \sigma$. If $p \neq q$, a second application of XI.4 would make l perpendicular to σ , although $l \subset \sigma$.

```

(hPerp :
  HilbertLinePerpendicularPlaneAt Geo l pi 0) :
Not (HilbertLineInPlane Geo l pi) := by

intro hlp

have hSelf :
  HilbertLinesPerpendicularAt Geo l l 0 :=
  HilbertLinePerpendicularPlaneAt.perpendicular_to_line
    (Geo := Geo)
    hPerp
    hlp
    hPerp.1

exact
  (hilbert_linesPerpendicularAt_ne
    Geo l l 0 hSelf) rfl

```

This helper is purely definitional in geometric content. If a line perpendicular to a plane were itself contained in the plane, then the universal line–plane perpendicularity condition would make the line perpendicular to itself. The predicate `HilbertLinesPerpendicularAt` is nondegenerate, so this is impossible.

The theorem body then consists of two plane constructions, one plane-intersection construction, and two calls to XI.4.

5.10 Hidden Formal Data

Several facts visible in a classical spatial diagram must be supplied explicitly in Lean.

- O lies in the plane determined by m, n .
- The ambient lines m, n must be packaged as `PlaneLine Geo pi`.
- Their distinctness must be transported to the subtype level.
- If $p \notin \pi$, the plane through l, p must be constructed explicitly.
- The two planes must be proved distinct before their intersection line can be used.
- The intersection line q must carry proofs that it lies in both π and σ .
- For the second XI.4 call, p, q and O are repackaged as `PlaneLine` and `PlanePoint` objects of σ .

These are mathematical incidence obligations, not merely syntactic casts. They encode information that Euclid’s diagram leaves implicit.

5.11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and XI.4.

The Lean proof imports Proposition XI.4 and uses the current spatial intersection theorem

```
hilbert_plane_intersection_line
```

instead of importing a separate formal Proposition XI.3.

Thus the two dependency graphs are not identical.

The actual Lean Book XI dependency is:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

Proposition XI.5 is then the coplanarity tool required by the classical route to Proposition XI.6.

5.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by the XI.5 theorem body.

The inherited planar Hilbert and Book Zero infrastructure enters transitively through Proposition XI.4. In particular, XI.4 internally uses induced plane geometry, while XI.5 itself is primarily a spatial incidence argument.

The new helper `hilbert_linePerpendicularPlaneAt_not_in_plane` does not add a geometric axiom; it follows from the existing definitions and nondegeneracy of line–line perpendicularity.

5.13 Axiomatic Status

The audit command is:

```
import CGJteamLab.Proposition11_5

#print axioms Geometry.euclid_proposition_11_5
```

The reported dependency set is:

```
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Therefore:

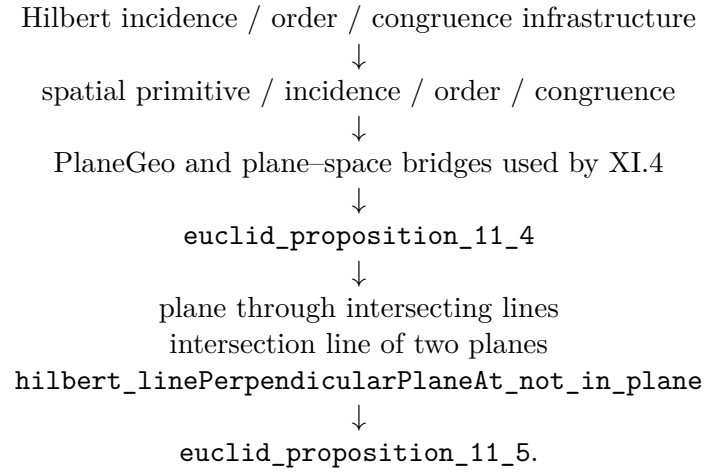
- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;

- new proposition-local helper: yes;
- new Hilbert3DAxioms declaration: no;
- new PlaneGeo bridge theorem: no;
- `sorry/admit`: no.

The inherited dependency `Geometry.bookZero_nullSegment1` is the same accepted dependency already present in Proposition XI.4.

5.14 Dependency Summary

The actual formal dependency chain is:



At the current checkpoint the production theorem compiles locally and its axiom audit has been run successfully. A full repository `lake build` is a separate checkpoint and should be recorded only after it is run.

5.15 Conclusion

Proposition XI.5 identifies the plane perpendicular to a given line at a given point as the unique carrier of all lines through that point perpendicular to the line, in the precise three-line form stated by Euclid.

The formal proof is shorter than the classical local proof graph because the present 3D interface already contains an intersection-line theorem for two planes and because XI.4 can be applied a second time to exclude the hypothetical second plane.

No coordinates, vectors, scalar products, continuity principle, or new spatial axiom enter the proof.

Chapter 6

Proposition XI.6

6.1 Introduction

Proposition XI.6 is the first proposition in the present Book XI development whose conclusion is a genuinely spatial parallelism statement. Two ambient lines l and m are perpendicular to the same plane π , at distinct feet B and D . Euclid concludes that the two lines are parallel.

In three-dimensional geometry this conclusion contains two logically different assertions. The lines must first be shown to be *coplanar*; only then can planar geometry prove that they do not meet. This distinction is essential in the formal reconstruction, because two disjoint ambient lines may be skew.

The current Lean proof follows the metric core of Euclid’s construction closely:

$$\text{auxiliary perpendicular and equal segment} \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5.}$$

After XI.5 has supplied the missing common plane, the proof changes deliberately to an induced planar geometry and closes the disjointness argument by the neutral alternate-angle criterion.

6.2 Euclid’s Statement

Euclid’s Proposition XI.6 states:

If two straight lines be at right angles to the same plane, the straight lines will be parallel.

In the classical configuration the two perpendiculars are AB and CD , meeting the reference plane at B and D . The formal theorem makes the nondegenerate case explicit by assuming

$$B \neq D.$$

6.3 Classical Configuration

Join the feet B and D . In the reference plane construct through D a line DE perpendicular to BD , and choose E so that

$$DE \cong AB.$$

Join BE , AE , and AD .

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through B , in particular to BD and BE . Hence

$$\angle ABD, \quad \angle ABE$$

are right angles.

Likewise CD is perpendicular to the reference plane at D , so it is perpendicular to both DB and DE .

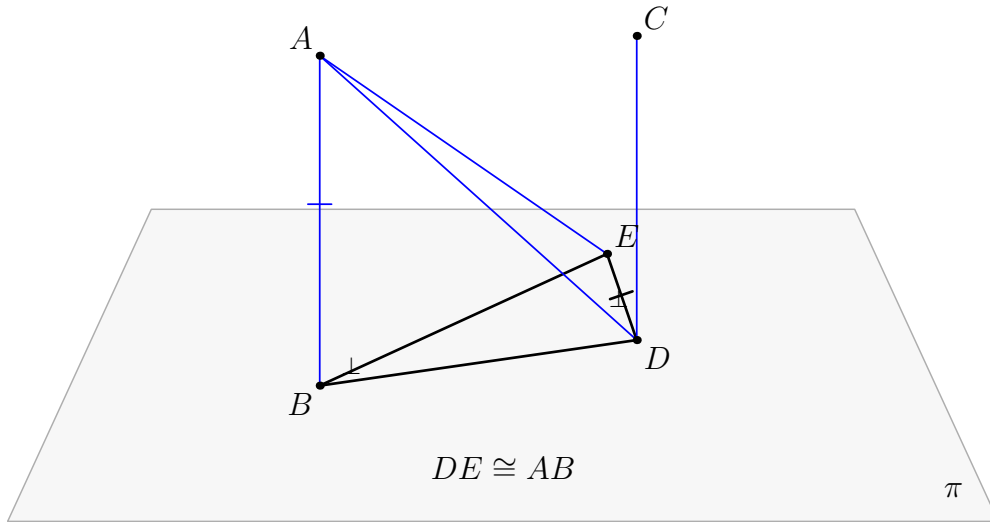


Figure 6.1: The classical configuration of Proposition XI.6. The black geometry lies in the reference plane π : B, D, E , the transversal BD , and the auxiliary perpendicular DE . The blue lines and connectors leave π . The construction $DE \cong AB$ prepares the SAS comparison of BAD and DEB ; the resulting equality $AD \cong EB$ prepares the SSS comparison of DEA and BAE .

6.4 Classical Proof

The classical metric part consists of two triangle-congruence steps.

First,

$$AB \cong DE, \quad BD \cong DB,$$

and the included angles

$$\angle ABD, \quad \angle EDB$$

are both right. Proposition I.4 therefore gives

$$AD \cong EB.$$

Second, the triangles DEA and BAE have

$$DE \cong BA, \quad EA \cong AE, \quad DA \cong BE.$$

Proposition I.8 gives

$$\angle EDA \cong \angle ABE.$$

Since $\angle ABE$ is right, $\angle EDA$ is right; hence

$$ED \perp DA.$$

Now at the point D the line ED is perpendicular to the three lines

$$DB, \quad DA, \quad DC.$$

Proposition XI.5 therefore places those three lines in one plane. Proposition XI.2 identifies the triangle plane determined by the intersecting lines, so AB is brought into the same plane as BD and DC . Finally

$$\angle ABD = \angle BDC = 90^\circ,$$

and Proposition I.28 yields

$$AB \parallel CD.$$

A compact classical dependency graph is therefore

$$\begin{array}{c}
 \text{construct } DE \perp BD, \quad DE \cong AB \\
 \Downarrow \\
 \text{I.4} \\
 \Downarrow \\
 AD \cong EB \\
 \Downarrow \\
 \text{I.8} \\
 \Downarrow \\
 ED \perp DA \\
 \Downarrow \\
 \text{XI.5} \\
 \Downarrow \\
 DB, DA, DC \text{ coplanar} \\
 \Downarrow \\
 \text{XI.2} \\
 \Downarrow \\
 AB, BD, DC \text{ coplanar} \\
 \Downarrow \\
 \text{I.28} \\
 \Downarrow \\
 AB \parallel CD.
 \end{array}$$

The construction of the auxiliary perpendicular and equal segment is part of the earlier Book I construction machinery. The substantive local congruence steps are I.4 and I.8.

6.5 What is genuinely three-dimensional here?

The difficult point is not the planar theorem that two perpendiculars to one transversal are parallel. That statement cannot even be applied until one knows that the two candidate parallel lines lie in one plane.

In ambient three-space the information

$$l \perp BD, \quad m \perp BD$$

does not by itself constitute a parallelism proof: without a coplanarity theorem the lines could still be skew.

The formal proof therefore separates XI.6 into

$$\boxed{\text{coplanarity}} \quad + \quad \boxed{\text{planar disjointness}}.$$

The first box is the genuinely spatial content and is exactly where Proposition XI.5 is used.

6.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_6
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence
    (Geo := Geo) (H := H) (S := S)]
  (pi : S.Plane)
  (l m : Geo.Line)
  (B D : Geo.Point)
  (hBD : Ne B D)
  (hLperp :
    HilbertLinePerpendicularPlaneAt
      Geo l pi B)
  (hMperp :
    HilbertLinePerpendicularPlaneAt
      Geo m pi D) :
  HilbertSpaceLinesParallel Geo l m

```

The target relation is part of the reusable spatial interface `Hilbert3DInterface`:

```

def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m

```

Thus “parallel in space” means exactly:

1. there is a plane containing both lines;
2. the two lines are disjoint.

The first clause is not redundant. If spatial parallelism were defined only by ambient disjointness, skew lines would incorrectly satisfy the definition.

6.7 Spatial / Planar Architecture Used

The proof uses three layers which must remain distinct.

6.7.1 Ambient three-space

The ambient objects and relations include

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

In particular, segment congruence between configurations carried by different planes is handled through `HilbertSpaceCongruence`.

6.7.2 Induced plane geometry

Whenever a planar theorem is needed, the relevant ambient plane σ or π is converted into

```
PlanePoint
PlaneLine
PlaneGeo
```

and ordinary planar incidence, order, right-angle, side-of-line, and triangle theorems are used there.

There is deliberately no global ambient instance

```
HilbertOrder Geo
HilbertCongruence Geo
```

for the three-dimensional space.

6.7.3 Plane-space bridges

The proof repeatedly uses bridges such as

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_sameRay_iff_ambient
```

These are not cosmetic casts. They state which geometric fact is being used in which plane slice.

6.8 Proof Architecture of the Reconstruction

6.8.1 Step 1: Euclid's auxiliary construction in the reference plane

Choose a point $A \neq B$ on the first normal l . Inside the reference plane π , construct the line

$$d = BD.$$

Through D construct a line e perpendicular to d , and lay off on e a point E such that

$$DE \cong AB.$$

This is packaged by

```
hilbert_XI6_auxiliary_perpendicular_equal_segment
```

The line d and the auxiliary line e are constructed in `PlaneGeo pi`. The segment copied onto DE , however, may have the spatial endpoint A ; that copy is therefore performed by `HilbertSpaceCongruence.segment_construction`.

6.8.2 Step 2: the first spatial SAS

Since $l \perp \pi$ at B and $d \subset \pi$ passes through B ,

$$l \perp d.$$

The auxiliary construction gives

$$e \perp d$$

at D .

The helper

```
hilbert_XI6_space_linesPerpendicularAt_right_angle_of_points
```

normalizes these line–line perpendicularities to the concrete right angles

$$\angle ABD, \quad \angle EDB.$$

The spatial right-angle comparison then supplies

$$\angle ABD \cong \angle EDB.$$

With

$$BA \cong DE, \quad BD \cong DB,$$

the spatial SAS theorem gives

$$AD \cong EB.$$

This whole block is packaged by

```
hilbert_XI6_first_SAS
```

and uses

```
hilbert_space_sas_third_side_and_angle
```

for the actual triangle comparison.

6.8.3 Step 3: the second spatial SSS and the new right angle

The next comparison is between the triangles DEA and BAE . The three side relations are

$$DE \cong BA, \quad EA \cong AE, \quad DA \cong BE.$$

The spatial SSS theorem gives

$$\angle EDA \cong \angle ABE.$$

Because $AB \perp \pi$ and $BE \subset \pi$, the angle ABE is right. The rightness is transported to EDA , yielding

$$ED \perp DA.$$

This block is packaged by

```
hilbert_XI6_second_SSS_right_angle
```

The proof must create explicit carrier planes for the relevant triangles. The right-angle transport is handled by

```
hilbert_XI6_space_right_angle_transport_in_plane
```

rather than by installing a forbidden ambient `HilbertCongruence Geo` instance.

6.8.4 Step 4: Proposition XI.5 supplies coplanarity

At D , the line $e = DE$ is now perpendicular to

$$d = DB, \quad a = DA, \quad m.$$

The last perpendicularity follows because $m \perp \pi$ and $e \subset \pi$.

Proposition XI.5 is applied exactly here:

```
euclid_proposition_11_5
```

It produces a plane σ containing

$$d, \quad a, \quad m.$$

Since $A \in a$ and $B \in d$, both A and B lie in σ . The line $l = AB$ therefore also lies in σ .

The result is isolated as

```
hilbert_XI6_normals_to_same_plane_coplanar
```

with conclusion

```
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma
```

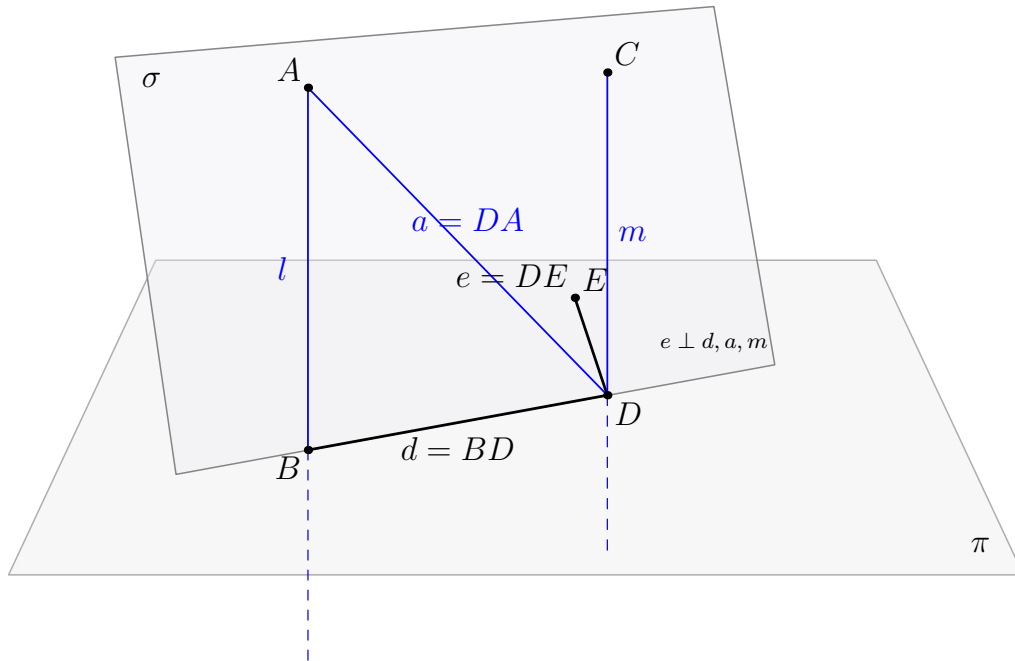


Figure 6.2: The genuinely spatial step in the formal XI.6 proof. The black lines $d = BD$ and $e = DE$ lie in the reference plane π . The blue lines $a = DA$, $m = DC$, and $l = AB$ leave π . At D , the line e is perpendicular to d, a, m , so XI.5 produces the second plane σ containing d, a, m . Because σ contains A and B , it also contains l . The lines l and m are therefore coplanar.

6.8.5 Step 5: the remaining problem is planar

After Step 4, both l and m lie in σ . The common transversal $d = BD$ lies in σ as well, and

$$l \perp d \text{ at } B, \quad m \perp d \text{ at } D.$$

The rest of the proof is carried out entirely inside

PlaneGeo Geo sigma.

The helper

```
hilbert_XI6_coplanar_normals_disjoint
```

chooses nonfoot points on l and m , establishes the required side-of-line orientation relative to d , compares the two right angles, and applies

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

to obtain planar parallelism.

The final conversion proves that the actual ambient carriers l and m are disjoint.

6.8.6 Step 6: assemble spatial parallelism

The public theorem now has exactly the two pieces required by `HilbertSpaceLinesParallel`:

$$l, m \subset \sigma$$

and

$$\text{HilbertLinesDisjoint}(l, m).$$

Therefore

$$\text{HilbertSpaceLinesParallel}(l, m).$$

6.9 Lean Formalization

The final theorem body is short because the geometric work has been isolated into two major helpers. First,

```
hilbert_XI6_normals_to_same_plane_coplanar
```

returns a plane `sigma` containing both ambient normals. Then

```
hilbert_XI6_coplanar_normals_disjoint
```

proves that their ambient carriers are disjoint. The public theorem packages exactly these three fields of `HilbertSpaceLinesParallel`: containment of l , containment of m , and disjointness.

The production module imports

```
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition27
```

The first import supplies the decisive spatial coplanarity theorem. The second supplies the neutral alternate-angle machinery used in the final induced plane.

6.10 Hidden Formal Data

The classical diagram suppresses several obligations which the Lean proof must make explicit.

- The feet B, D are distinct.
- The point A selected on l is outside the reference plane π .
- The auxiliary point E is nondegenerate and lies both on e and in π .
- Every triangle used in SAS or SSS carries an explicit noncollinearity proof.
- Spatial triangle comparisons require an explicit target carrier plane.
- Right-angle transport between ambient configurations is performed through an explicit `PlaneGeo`, not through an ambient planar congruence instance.
- Before XI.5 can be applied, the three lines d, a, m must be proved distinct in the required places and must all pass through D .
- The plane σ returned by XI.5 must be shown to contain both A and B , so that the whole carrier $l = AB$ lies in σ .
- In the final planar proof, the chosen points on l and m must be oriented on opposite sides of the transversal d . The proof performs an explicit `SameSide/OppositeSide` case split.
- Point-pair parallelism inside `PlaneGeo` `sigma` must finally be converted into disjointness of the original ambient carriers.

These obligations are geometric. They are the incidence, order, and plane-membership data that a conventional three-dimensional drawing usually leaves implicit.

6.11 Relationship with Earlier Book XI Propositions

Proposition XI.5 is a direct formal dependency:

$$\boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

XI.5 itself depends on XI.4, and XI.6 also reuses several XI.4 normalization helpers for perpendicular lines and right angles. Consequently the effective Book XI chain is

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

The classical proof also invokes XI.2 when it moves from the three coplanar lines through D to the plane containing the triangle ABD . The Lean proof does not need a separate formal call to Proposition XI.2: the spatial incidence API directly constructs and tracks the relevant carrier plane.

6.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The metric part of the formal proof reproduces the roles of Euclid I.4 and I.8 through the spatial SAS and SSS infrastructure:


```
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

Right angles and their transport are handled synthetically; no angle measure is introduced.

The classical final step is I.28. The production Lean file instead imports `Proposition27` and uses the established neutral alternate-angle theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

inside the explicitly constructed plane σ . Thus the formal module dependency and the classical Euclid proposition-number dependency are not identical, even though they close the same planar configuration.

Book Zero enters only transitively through the established Hilbert geometric infrastructure. No new Book Zero axiom or theorem is added by XI.6.

6.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_6

#print axioms Geometry.euclid_proposition_11_6
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same reported axiom profile as Propositions XI.4 and XI.5.

Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;
- new proposition-local helper theorems: yes;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` declaration: `HilbertSpaceLinesParallel`, promoted from the XI.6 proposition module for reuse by XI.7 and later spatial results;

- new PlaneGeo bridge theorem: no;

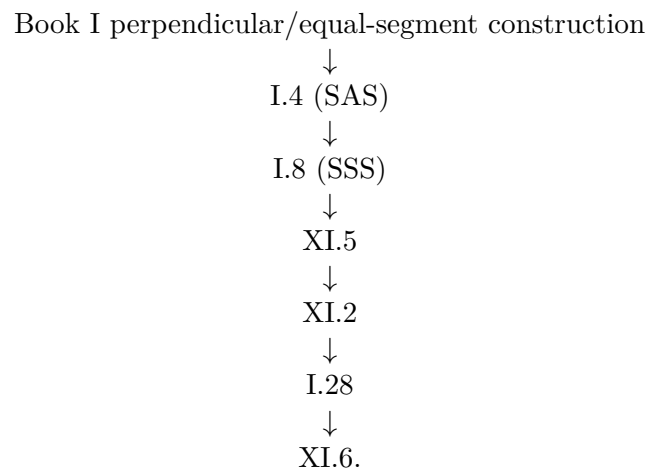
- sorry/admit: no.

The inherited `Geometry.bookZero_nullSegment1` dependency is not specific to XI.6.

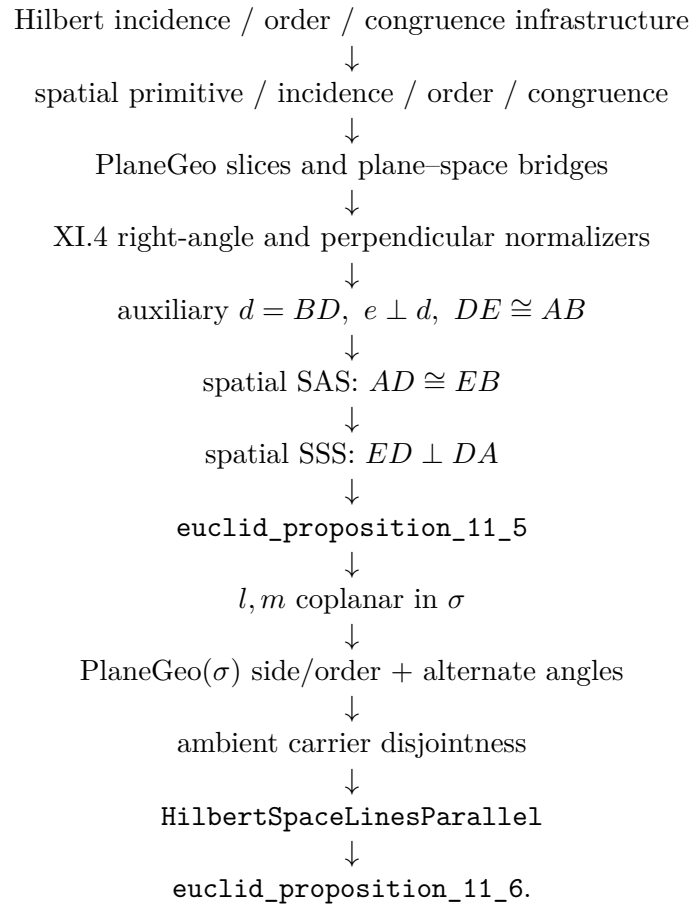
The production theorem compiled successfully, its axiom audit was run successfully, and the full repository `lake build` was reported clean at this checkpoint.

6.14 Dependency Summary

The classical local chain is



The actual formal chain is



6.15 Conclusion

Proposition XI.6 is a useful boundary theorem for the three-dimensional architecture. Its metric core is recognizably Euclidean: construct an auxiliary perpendicular and an equal segment, use SAS, then SSS, and obtain a third perpendicular at D .

The genuinely spatial step is XI.5. It converts three perpendicularities through one point into the common plane in which the final parallelism argument is allowed to take place. Lean makes this logical boundary explicit rather than allowing the diagram to hide it.

The final definition of spatial parallelism records the same distinction: coplanarity is proved first and disjointness second. No coordinates, vectors, scalar products, numerical angle measure, continuity principle, new spatial axiom, or Euclidean parallel axiom is introduced.

Chapter 7

Proposition XI.7

7.1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert’s spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid’s *reductio*. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

7.2 Euclid’s Statement

Euclid’s Proposition XI.7 states:

If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD . The claim is that the joining straight line EF lies in the plane of AB and CD .

7.3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD . The points E and F are arbitrary points on the two lines.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

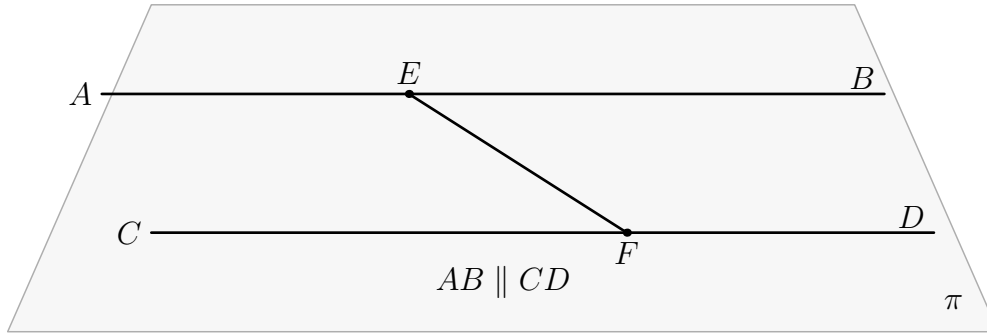


Figure 7.1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π , with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π . All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

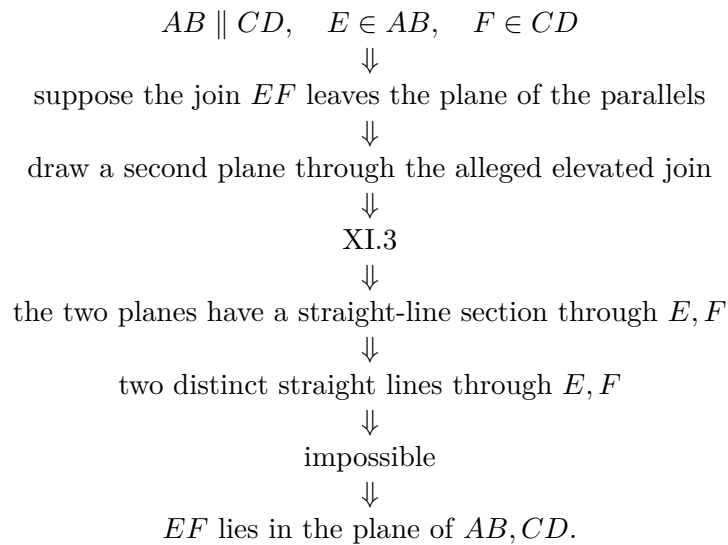
7.4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F . Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore



The only explicit Book XI proposition cited in the proof is XI.3.

7.5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

coplanarity and disjointness.

XI.7 uses the coplanarity component directly. No metric geometry is needed.

7.6 Formal Statement

The production theorem is:

```

theorem euclid_proposition_11_7
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  (l m : Geo.Line)
  (E F : Geo.Point)
  (hParallel : HilbertSpaceLinesParallel Geo l m)
  (hEl : H.OnLine E l)
  (hFm : H.OnLine F m) :
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
  exists n : Geo.Line,
    H.OnLine E n /\
    H.OnLine F n /\
    HilbertLineInPlane Geo n sigma

```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through `E` and `F`, together with the proof that `n` lies in `sigma`.

7.7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```

HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel

```

No induced **PlaneGeo** is constructed. There is no planar betweenness, SameRay, SameSide/OppositeSide, angle congruence, right angle, SAS, SSS, or plane-space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

7.8 Proof Architecture of the Reconstruction

7.8.1 Step 1: unpack spatial parallelism

From

$$\text{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \quad m \subset \sigma,$$

together with

$$\text{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general **Hilbert3DInterface** rather than a proposition-local declaration.

7.8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

7.8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid's diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m , contradicting

$$\text{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

7.8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with

$$E \in n, \quad F \in n.$$

7.8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n , and both lie in σ . The decisive step is:

`HilbertSpaceIncidence.line_in_plane`

which gives

$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert’s incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

7.9 Lean Formalization

The proof DAG is extremely short:

$$\begin{array}{c}
 \text{HilbertSpaceLinesParallel } l \ m \\
 \Downarrow \\
 \exists \sigma, \quad l \subset \sigma, \quad m \subset \sigma, \quad l \cap m = \emptyset \\
 \Downarrow \\
 E \in \sigma, \quad F \in \sigma \\
 \Downarrow \\
 E \neq F \\
 \Downarrow \\
 \text{line_through } E \ F \\
 \Downarrow \\
 n \text{ through } E, F \\
 \Downarrow \\
 \text{HilbertSpaceIncidence.line_in_plane} \\
 \Downarrow \\
 n \subset \sigma \\
 \Downarrow \\
 \text{euclid_proposition_11_7.}
 \end{array}$$

There are no proposition-local helper theorems.

7.10 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase “the straight line joining E and F ” is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

7.11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. Hilbert’s incidence system takes the closure of a line under two plane points as an explicit spatial incidence axiom.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

7.12 Relationship with Book I / Hilbert Infrastructure

Euclid’s definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive Hilbert source is Group I, Axiom I.6. In Hilbert’s classification, I.1–I.3 are the plane incidence axioms and I.4–I.8 are the space incidence axioms. Thus the formal proof lands exactly in the spatial incidence layer and nowhere higher in the hierarchy.

7.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports:

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This statement must be interpreted precisely. The theorem still has explicit typeclass hypotheses, including `HilbertSpaceIncidence Geo`. The audit says that its proof term does not depend on any additional global Lean axioms or project axiom constants beyond those explicit parameters.

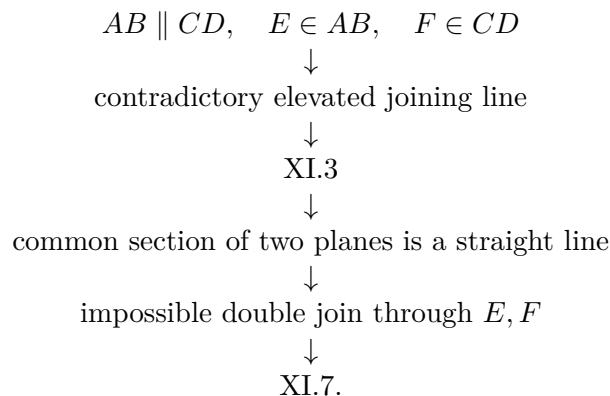
Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;
- spatial order: not used;
- spatial congruence: not used;
- induced `PlaneGeo`: not used;
- new proposition-local helper theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` change: yes, `HilbertSpaceLinesParallel` was promoted there;
- new `PlaneGeo` bridge theorem: no;
- `sorry/admit`: no.

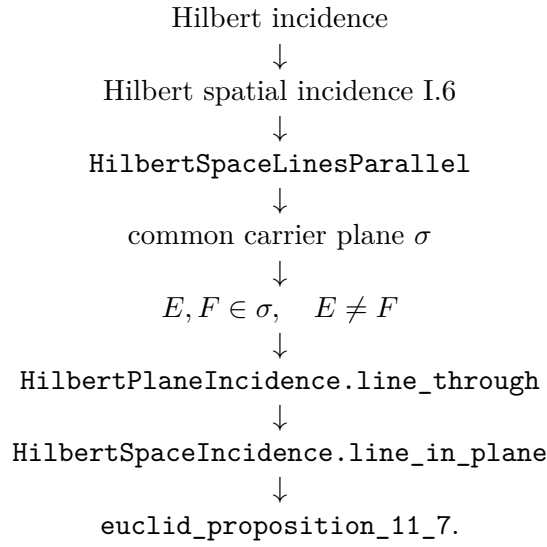
The production theorem and the full repository build were reported clean at this checkpoint.

7.14 Dependency Summary

The classical local chain is



The actual formal chain is



7.15 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The Hilbert reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and Axiom I.6 says that a line with two distinct points in that plane lies wholly in the plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.

Chapter 8

Proposition XI.8

8.1 Introduction

Proposition XI.8 is the Euclidean converse companion to XI.6. Proposition XI.6 proved that two straight lines perpendicular to the same plane are parallel. Proposition XI.8 starts with two parallel spatial lines and assumes that one of them is perpendicular to a reference plane; the conclusion is that the second parallel is perpendicular to the same plane.

The proposition is also a structural boundary in the reconstruction. The formalizations XI.4–XI.7 did not require Hilbert Group IV. XI.8 does. The reason is already visible in Euclid’s classical proof: the decisive transversal step uses Proposition I.29, the converse direction of the neutral parallel-angle criterion.

The Lean proof preserves the synthetic geometry but makes the Euclidean dependency more explicit. Instead of calling a planar I.29 wrapper, it separates the argument into

neutral I.27 + uniqueness of the parallel through a point.

The second component is precisely the plane-local Group IV assumption.

A second feature is genuinely three-dimensional. Euclid begins by saying that the two parallel lines meet the reference plane at points B and D . The first intersection is given by the hypothesis that the first line is perpendicular to the plane. The second intersection is not automatic in an abstract incidence space. The formal proof derives it.

8.2 Euclid’s Statement

Euclid’s Proposition XI.8 states:

If two straight lines be parallel, and one of them be at right angles to any plane, the remaining one will also be at right angles to the same plane.

Let AB and CD be parallel straight lines, and suppose that AB is perpendicular to the reference plane π at B . The claim is that CD meets π at a point D and is perpendicular to π there.

8.3 Classical Configuration

Euclid lets the parallel lines AB and CD meet the reference plane at B and D . He joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

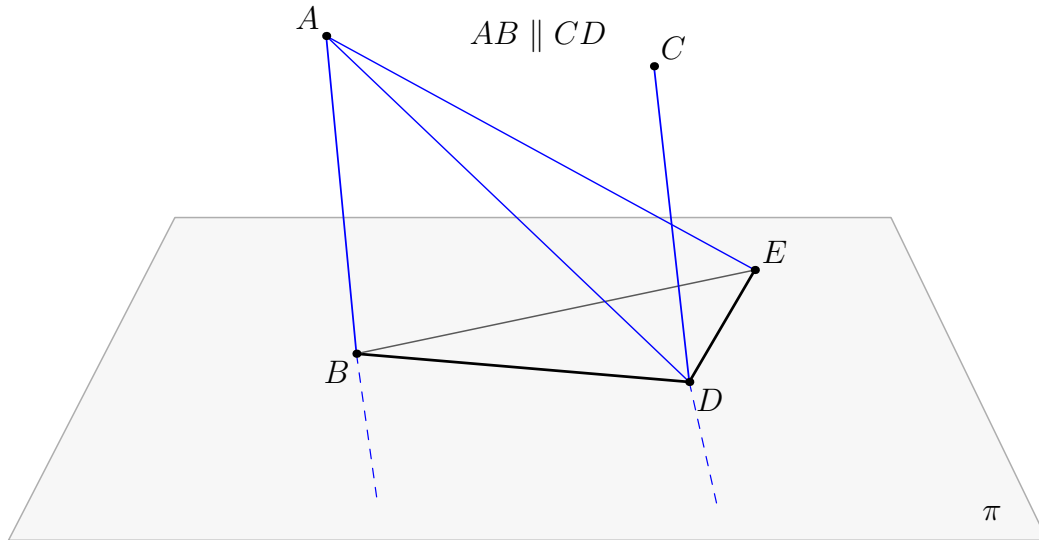


Figure 8.1: The classical configuration of Proposition XI.8. Black lines lie in the reference plane π . Blue lines or segments leave π . The hidden continuations of the two spatial parallel carriers are shown blue and dashed. The auxiliary construction reproduces the metric core already used in XI.6: $DE \perp BD$ and $DE = AB$.

The line BD plays two roles. It is a line of the reference plane and also lies in the common plane of the spatial parallels. This is the transversal on which the classical I.29 step acts.

8.4 Classical Proof

The local classical proof has five main stages.

First, since $AB \parallel CD$, Proposition XI.7 places AB , CD , and the joining line BD in one plane. The opening phrase that the parallel lines meet the reference plane at B, D is diagrammatic in the classical text.

Second, Euclid constructs DE in the reference plane with

$$DE \perp BD, \quad DE = AB.$$

Since $AB \perp \pi$, the angles

$$\angle ABD, \quad \angle ABE$$

are right.

Third, BD falls on the parallel lines AB and CD . Proposition I.29 says that the appropriate interior angles sum to two right angles. Since $\angle ABD$ is right, $\angle CDB$ is right. Hence

$$CD \perp BD.$$

Fourth, Euclid repeats the metric triangle argument familiar from XI.6. From

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles, Proposition I.4 gives

$$AD = BE.$$

Then

$$AB = DE, \quad BE = AD, \quad AE = EA,$$

so Proposition I.8 gives

$$\angle ABE = \angle EDA.$$

The former is right, hence the latter is right:

$$ED \perp AD.$$

Now ED is perpendicular to the two intersecting lines DB and DA . Proposition XI.4 therefore gives

$$ED \perp \text{plane}(BDA).$$

The line CD lies in this plane, so

$$ED \perp CD.$$

Finally, in the reference plane π the two lines DB and DE meet at D , and CD is perpendicular to both:

$$CD \perp DB, \quad CD \perp DE.$$

A second application of XI.4 yields

$$CD \perp \pi.$$

The classical local dependency chain is therefore

$$\begin{array}{c}
 AB \parallel CD, \quad AB \perp \pi \\
 \Downarrow \\
 \text{XI.7} \quad \text{and the joining line } BD \\
 \Downarrow \\
 DE \perp BD, \quad DE = AB \quad (\text{I.11, I.3}) \\
 \Downarrow \\
 AB \perp \pi \Rightarrow \angle ABD, \angle ABE \text{ right} \\
 \Downarrow \\
 \text{I.29} \\
 \Downarrow \\
 CD \perp BD \\
 \Downarrow \\
 \text{I.4, I.8} \\
 \Downarrow \\
 ED \perp AD \\
 \Downarrow \\
 \text{XI.4} \\
 \Downarrow \\
 ED \perp CD \\
 \Downarrow \\
 \text{XI.4} \\
 \Downarrow \\
 CD \perp \pi.
 \end{array}$$

The incidence assertion that CD belongs to the plane through BD, DA also uses the fact that the triangle ABD is planar and that AB, CD, BD share the plane supplied by XI.7. In a formal proof this plane identification cannot be left to the printed diagram.

8.5 What is genuinely three-dimensional here?

There are two distinguished planes.

The first is the reference plane π , to which AB is perpendicular. The second is the common plane σ containing the parallel spatial lines AB and CD . Since AB is not contained in π , the planes are distinct.

The points B and D lie in both planes. Consequently their joining line

$$d = BD$$

lies in both π and σ ; in the formal proof it is the relevant line of intersection of the two plane slices.

The auxiliary line

$$e = DE$$

lies in π , while the line

$$a = DA$$

lies in σ . The proof therefore alternates between two induced plane geometries.

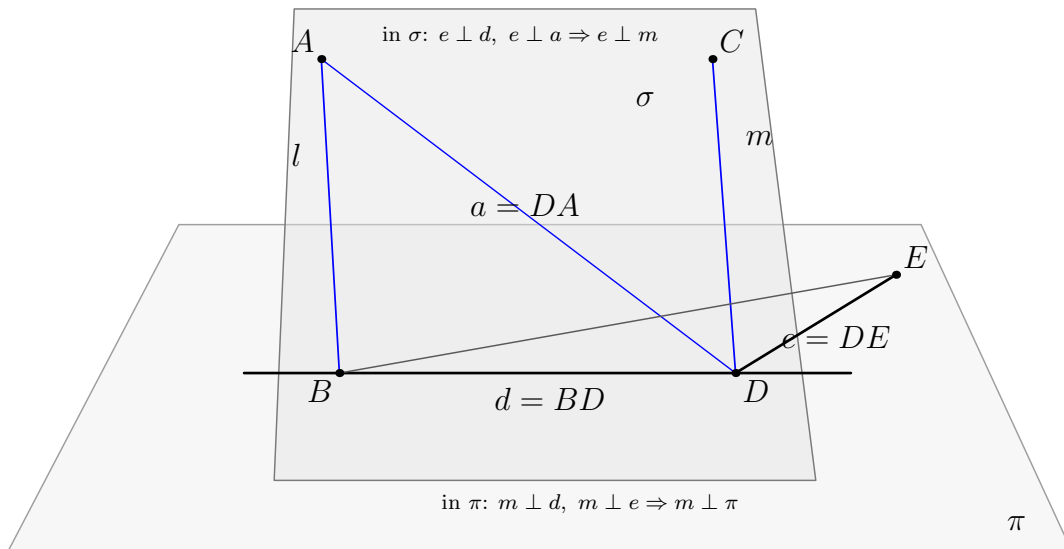


Figure 8.2: The two plane slices used by the reconstruction. The common line $d = BD$ lies in both π and σ . The first XI.4 application uses the two lines $d, a \subset \sigma$ to prove $e \perp \sigma$, hence $e \perp m$. The final XI.4 application uses $d, e \subset \pi$ to prove $m \perp \pi$.

This change of plane is not bookkeeping. It is the spatial geometry of the proposition.

8.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_8
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
```

```

[HSI : HilbertSpaceIncidence Geo]
[_HSO : HilbertSpaceOrder
  (Geo := Geo) (H := H) (S := S)]
[HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(l m : Geo.Line)
(pi : S.Plane)
(B : Geo.Point)
(hParallel : HilbertSpaceLinesParallel Geo l m)
(hPerp : HilbertLinePerpendicularPlaneAt Geo l pi B) :
exists D : Geo.Point,
  HilbertLinePerpendicularPlaneAt Geo m pi D

```

The point D is deliberately not an input. Its existence is part of the theorem. This is stronger as a formal interface than assuming in advance that the second parallel already meets the plane.

8.7 Spatial / Planar Architecture Used

The ambient spatial layer consists of

```

HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertSpaceEuclidean
HilbertSpaceLinesParallel
HilbertLinePerpendicularPlaneAt
HilbertLinesPerpendicularAt

```

The proof then uses induced planar geometry in two different fixed planes:

$$\text{PlaneGeo}(Geo, \sigma) \quad \text{and} \quad \text{PlaneGeo}(Geo, \pi).$$

Inside σ , the proof uses neutral planar order, right angles, and the I.27 alternate-angle criterion. Inside π , the final configuration supplies two distinct intersecting lines d, e for XI.4.

The plane-space bridges used in the argument include

```

planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_linesDisjoint_iff_ambient

```

The last bridge was added with the Group IV layer. It identifies disjointness of two lines in a fixed **PlaneGeo** with ambient disjointness of their underlying spatial carriers.

8.8 The Euclidean Layer: Group IV

XI.8 is the first proposition in the present Book XI sequence whose formal proof requires the Euclidean parallel axiom.

The spatial axiom is not installed as a global `HilbertEuclideanPlaneGeo` instance. Such an instance would incorrectly treat the ambient three-space as a plane. Instead the axiom is stated plane-locally:

```
class HilbertSpaceEuclidean
  (Geo : Geometry.Geo)
  [H : HilbertIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HilbertSpaceIncidence Geo]
  [HilbertSpaceOrder Geo]
  [HilbertSpaceCongruence Geo] : Prop where

parallel_unique_in_plane :
  forall pi : S.Plane,
    forall l : Geo.Line,
      HilbertLineInPlane Geo l pi ->
        forall A : Geo.Point,
          S.OnPlane A pi ->
            Not (H.OnLine A l) ->
              forall b c : Geo.Line,
                HilbertLineInPlane Geo b pi ->
                  HilbertLineInPlane Geo c pi ->
                    H.OnLine A b ->
                      HilbertLinesDisjoint Geo b l ->
                        H.OnLine A c ->
                          HilbertLinesDisjoint Geo c l ->
                            b = c
```

For every fixed ambient plane, the bridge instance

```
instance planeGeoHilbertEuclidean
  ...
  (pi : S.Plane) :
    HilbertEuclideanPlane (PlaneGeo Geo pi)
```

recovers the existing planar Group IV API. The architectural rule is therefore preserved:

Euclidean parallelism is planar, even inside three-space.

8.9 Proof Architecture of the Reconstruction

8.9.1 Step 1: derive the hidden intersection point D

The helper

```
hilbert_XI8_parallel_meets_perpendicular_plane
```

proves that the second parallel m meets the reference plane.

Unpacking

$$\text{HilbertSpaceLinesParallel}(l, m)$$

gives a common plane σ containing l, m , with the two lines disjoint. The foot B of $l \perp \pi$ lies in both π and σ .

The planes are distinct. If $\sigma = \pi$, then $l \subset \pi$, contradicting the theorem

```
hilbert_linePerpendicularPlaneAt_not_in_plane
```

The general plane-intersection theorem then supplies a line

$$q = \pi \cap \sigma$$

through B .

Suppose m did not meet q . Then both l and q would be lines through B in σ disjoint from m . Group IV uniqueness would force $l = q$, but $q \subset \pi$, again contradicting $l \not\subset \pi$.

Hence m meets q at some point D , and therefore

$$D \in m \cap \pi.$$

This is the formal proof of a fact Euclid leaves implicit at the opening of XI.8.

8.9.2 Step 2: reuse the XI.6 metric core

With D fixed, the proof reuses

```
hilbert_XI6_second_SSS_right_angle
```

from Proposition XI.6.

The helper constructs points and lines corresponding to Euclid's auxiliary configuration and returns, in particular,

$$d = BD, \quad e = DE,$$

with

$$d, e \subset \pi, \quad e \perp d,$$

and the right angle

$$\angle EDA.$$

Thus the SAS/SSS part of XI.8 is not duplicated. The earlier XI.6 development is used as reusable spatial congruence infrastructure.

8.9.3 Step 3: first application of XI.4 in σ

Let a be the line DA . The proof shows

$$d, a \subset \sigma, \quad d \neq a.$$

The inequality is geometrically meaningful: A lies on the spatial line l but outside the reference plane π , while $d \subset \pi$.

The XI.6 metric result gives

$$e \perp d, \quad e \perp a.$$

Therefore XI.4, applied with the induced geometry of σ , gives

$$e \perp \sigma.$$

Since $m \subset \sigma$ and $D \in m$,

$$e \perp m.$$

This is the formal counterpart of Euclid's first XI.4 step.

8.9.4 Step 4: replace I.29 by neutral I.27 plus Group IV

The remaining task is to prove

$$m \perp d.$$

The proof does not call a public I.29 theorem. Instead, inside $\text{PlaneGeo}(\text{Geo}, \sigma)$, it constructs a line p through D perpendicular to d .

The neutral helper

```
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
```

shows that p and l are disjoint. Its proof is an I.27 argument: two right angles are normalized to equal alternate angles and the neutral alternate-angle theorem produces nonintersection.

But m is also disjoint from l , because $l \parallel m$. Thus m and p are two lines of the same plane σ , both passing through D , both disjoint from l . Group IV uniqueness forces

$$m = p.$$

Since $p \perp d$,

$$m \perp d.$$

This factorization isolates the exact Euclidean content:

$$\underbrace{\text{right angles} \Rightarrow \text{nonintersection}}_{\text{neutral I.27}} + \underbrace{\text{unique parallel through } D}_{\text{Group IV}} \implies m \perp d.$$

It is the formal counterpart of Euclid's I.29 step.

8.9.5 Step 5: final application of XI.4 in π

At this point

$$d, e \subset \pi, \quad m \perp d, \quad m \perp e.$$

The lines d and e are distinct because $e \perp d$.

The final call to

```
euclid_proposition_11_4
```

uses d, e as the two distinct intersecting lines of the reference plane. It returns

$$m \perp \pi$$

at D , which is the required conclusion.

8.10 Lean Formalization

The proposition-local proof is organized around four named helpers:

```
hilbert_XI8_parallel_meets_perpendicular_plane
hilbert_XI8_auxiliary_line_perpendicular_to_parallel
hilbert_XI8_coplanar_perpendiculars_to_same_line_disjoint
hilbert_XI8_parallel_line_perpendicular_to_given_BD
```

The actual Lean DAG is

$$\begin{array}{c}
 l \parallel m, \quad l \perp \pi \text{ at } B \\
 \Downarrow \\
 \text{hilbert_XI8_parallel_meets_perpendicular_plane} \\
 \Downarrow \\
 D \in m \cap \pi \\
 \Downarrow \\
 \text{hilbert_XI6_second_SSS_right_angle} \\
 \Downarrow \\
 d = BD, \quad e = DE, \quad e \perp d, \quad \angle EDA \text{ right} \\
 \Downarrow \\
 \text{XI.4 in } \sigma \\
 \Downarrow \\
 e \perp m \\
 \Downarrow \\
 \text{I.27 in PlaneGeo}(\sigma) + \text{Group IV} \\
 \Downarrow \\
 m \perp d \\
 \Downarrow \\
 \text{XI.4 in } \pi \\
 \Downarrow \\
 \text{euclid_proposition_11_8.}
 \end{array}$$

The production file imports XI.4, XI.5, and XI.6. It does not import XI.7. The common plane that Euclid obtains through XI.7 is already contained in the formal hypothesis `HilbertSpaceLinesParallel`.

8.11 Hidden Formal Data

XI.8 contains substantially more hidden incidence information than its classical diagram suggests.

- The second parallel must be proved to meet the reference plane. The point D is an existential output, not an assumed input.
- The common plane σ of the parallels must be proved distinct from π .
- The intersection line $q = \pi \cap \sigma$ is constructed explicitly.
- The points B, D must be proved distinct from disjointness of the parallel carriers.
- The line $d = BD$ must be shown to lie both in π and in σ .
- The point A selected on l must be proved to lie outside π ; otherwise the line l would lie in π , contrary to $l \perp \pi$.
- The line $a = DA$ must be constructed and shown distinct from d .
- Right-angle and perpendicularity statements must be transported between ambient geometry and PlaneGeo.
- The neutral I.27 argument requires explicit SameSide / OppositeSide and SameRay normalization that Euclid suppresses in the diagram.
- The two disjointness orientations required by Group IV are supplied explicitly.
- Before the final XI.4 application, $d \neq e$ is derived from $e \perp d$.

Subtype projections and record fields occur in the Lean code, but they are implementation details. The mathematically significant hidden data are the plane memberships, distinctness statements, side orientations, and the derived intersection point D .

8.12 Relationship with Earlier Book XI Propositions

The classical and formal dependency graphs differ in a controlled way.

Classically, XI.7 is an explicit opening dependency: it places AB, CD, BD in one plane. XI.4 is then used twice. The classical plane-identification step also relies on the ordinary fact from XI.2 that a triangle lies in one plane.

The production Lean file imports

```
import CGJteamLab.Proposition11_4
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition11_6
```

but does not import Proposition XI.7.

XI.4 is used twice exactly as in the classical architecture: first in the common plane σ , then in the reference plane π .

XI.5 contributes established line–plane perpendicularity infrastructure, including the theorem that a line perpendicular to a plane is not contained in that plane.

XI.6 contributes the reusable metric construction and the spatial SAS/SSS right-angle machinery. This reuse is substantial: the central auxiliary configuration of XI.8 is almost the same as the configuration built for XI.6.

XI.7 is absent from the Lean import graph because `HilbertSpaceLinesParallel` already stores the common carrier plane as part of the definition of spatial parallelism. Thus the formal proof consumes the conclusion-like data that XI.7 exposes, without invoking the proposition theorem itself.

8.13 Relationship with Book I / Book Zero / Hilbert Infrastructure

The classical proof uses I.11 and I.3 to construct the auxiliary perpendicular and equal segment. In the Lean reconstruction these operations are already packaged in the XI.6 helper chain over the Hilbert/Book Zero construction interface.

The metric part corresponds to I.4 and I.8 and is likewise inherited from XI.6.

The parallel step is the important logical boundary. The direction represented by I.27,

$$\text{equal alternate angles} \implies \text{nonintersection},$$

belongs to the neutral congruence/order layer. The reverse use of parallelism represented by I.29 requires the Euclidean parallel axiom. The formal proof displays this by using the neutral I.27 theorem first and applying Group IV only to identify the unique line through D disjoint from l .

No coordinate system, vector space, scalar product, distance formula, trigonometric function, or numerical angle measure is used.

8.14 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_8

#print axioms Geometry.euclid_proposition_11_8
```

and Lean reports:

```
'Geometry.euclid_proposition_11_8' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited dependencies already accepted in the Book XI development. XI.8 introduces no new proposition-specific global Lean axiom constant.

The Group IV assumption must be distinguished from this audit output. The public theorem explicitly assumes

```
[HSE : HilbertSpaceEuclidean Geo]
```

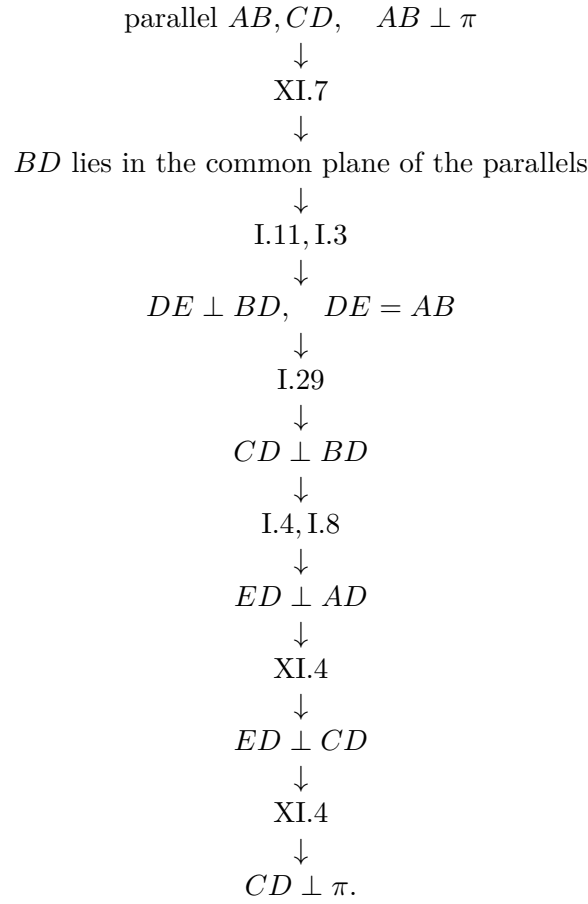
as part of its geometric hypothesis. This typeclass represents the Euclidean parallel axiom for spatial geometry, stated plane-locally. It is therefore a genuine mathematical assumption of XI.8 even though it does not appear as an additional global constant in `#print axioms`.

At this checkpoint:

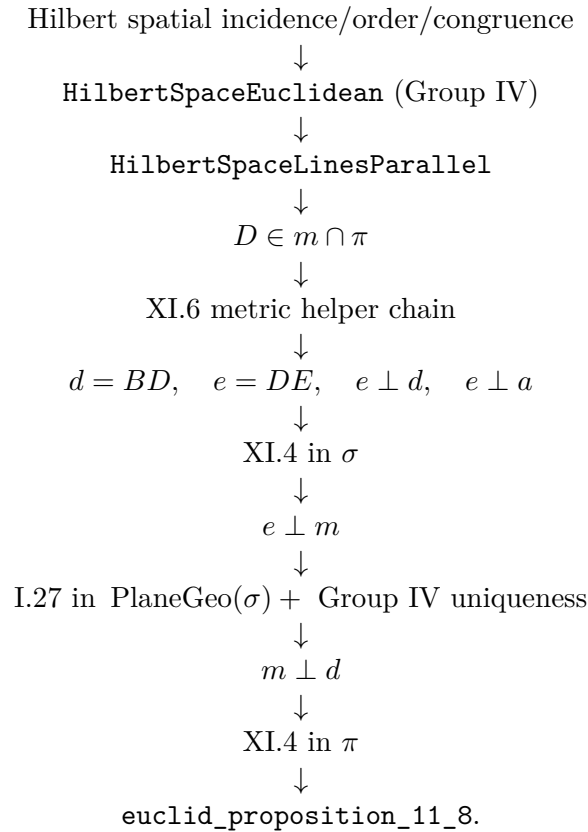
- new proposition-specific geometric axiom: no;
- new spatial axiom declaration in the general architecture: yes, `HilbertSpaceEuclidean`;
- Euclidean parallel axiom / Hilbert Group IV: used;
- new continuity assumption: no;
- new proposition-local helper theorems: yes;
- new reusable `Hilbert3DInterface` theorem: yes, `planeGeo_linesDisjoint_iff_ambient`;
- new induced-plane Euclidean bridge: yes, `planeGeoHilbertEuclidean`;
- global ambient `HilbertEuclideanPlaneGeo`: not installed;
- public theorem compiles: yes, as reported at this checkpoint;
- full repository build: clean, as reported at this checkpoint;
- `sorry/admit`: no.

8.15 Dependency Summary

The classical local chain is



The actual formal chain is



The dependency classification is therefore:

New proposition-specific global axiom	no
New spatial axiom class	yes: HilbertSpaceEuclidean
Continuity	no
Group IV / Euclidean parallelism	yes
Earlier Book XI production imports	XI.4, XI.5, XI.6
Classical explicit Book XI dependency	XI.7, XI.4
New PlaneGeo Euclidean bridge	yes
Coordinates / vectors / scalar product	no
sorry/admit	no

8.16 Conclusion

Proposition XI.8 is the first point in the present Book XI reconstruction where Euclidean parallel uniqueness is mathematically necessary. The formal proof makes that boundary sharper than the classical text: neutral I.27 proves nonintersection, while Group IV supplies the uniqueness needed to recover the I.29 direction.

At the same time, the proof remains fully synthetic. It reuses the metric construction of XI.6, applies XI.4 in two different plane slices, and derives the hidden intersection point of the second parallel with the reference plane instead of assuming it from the diagram.

The result therefore tests three parts of the spatial architecture at once: plane-local Euclidean parallelism, transport between ambient three-space and induced plane geometry, and the universal line–plane perpendicularity predicate. No continuity or analytic representation is introduced.

Appendix A

Euclid in Heath's Edition: Source Audit

A.1 Scope

This appendix records the Book XI source audit against Euclid's *Elements* in Heath's edition through Proposition XI.8 [1]. The purpose is to control the classical statement, construction, and local proposition dependencies. Lean-specific incidence, subtype, and plane-transport obligations are documented in the proposition chapters and are not attributed to Euclid or Heath.

A.2 Proposition XI.4

Proposition XI.4 states that if a straight line is set up at right angles to each of two straight lines at their point of intersection, then it is also at right angles to the plane through those two lines.

The classical proof chooses equal segments on the two given lines and introduces an arbitrary line of the plane through the common point. Triangle-congruence arguments are then used to prove that the candidate spatial line makes a right angle with that arbitrary transversal.

For the formal reconstruction the significant classical features are:

- the target is universal over every line of the plane through the common point;
- the arbitrary transversal is part of the proposition's logical content, not merely a feature of the diagram;
- the proof is synthetic and based on incidence, right angles, equal segments, and triangle congruence.

A.3 Proposition XI.5

Proposition XI.5 says that if one straight line is perpendicular at a common point to three straight lines meeting there, then those three straight lines lie in one plane.

Euclid's contradiction argument uses the plane through two of the three candidate coplanar lines and another plane through the common perpendicular and the remaining line. The intersection

line of the two planes produces, inside one plane, two distinct lines through the same point both perpendicular to the common perpendicular.

The local classical dependency is therefore governed by the preceding spatial propositions: the intersection-of-planes configuration and Proposition XI.4 supply the contradiction. The current Lean proof organizes this material differently, but preserves the same spatial core: perpendicularity to a plane forces the required intersection line to be perpendicular to the common normal.

A.4 Proposition XI.6

Heath gives the statement:

If two straight lines be at right angles to the same plane, the straight lines will be parallel.

Let AB and CD be perpendicular to the reference plane at B and D . Euclid joins BD , draws DE in the reference plane perpendicular to BD , makes

$$DE = AB,$$

and joins BE , AE , and AD .

Since AB is perpendicular to the reference plane,

$$\angle ABD, \quad \angle ABE$$

are right. Since CD is also perpendicular to the same plane, the corresponding lines of the plane through D are perpendicular to CD .

The first triangle comparison uses

$$AB = DE, \quad BD = DB,$$

and equality of the included right angles. Proposition I.4 gives

$$AD = BE.$$

The second comparison uses

$$DE = BA, \quad EA = AE, \quad DA = BE.$$

Proposition I.8 therefore gives

$$\angle EDA = \angle ABE.$$

Hence $\angle EDA$ is right and

$$ED \perp DA.$$

At D , the line ED is now perpendicular to

$$DB, \quad DA, \quad DC.$$

Proposition XI.5 places these three lines in one plane. Proposition XI.2 supplies the corresponding triangle-plane incidence, bringing AB into that same plane. The two right angles made with the transversal BD then give the final parallelism by Proposition I.28.

The local classical DAG can be recorded as

$$\begin{array}{c}
\text{construct } DE \perp BD, \quad DE = AB \\
\Downarrow \\
\text{I.4} \\
\Downarrow \\
AD = BE \\
\Downarrow \\
\text{I.8} \\
\Downarrow \\
ED \perp DA \\
\Downarrow \\
\text{XI.5} \\
\Downarrow \\
DB, DA, DC \text{ coplanar} \\
\Downarrow \\
\text{XI.2} \\
\Downarrow \\
AB, BD, DC \text{ coplanar} \\
\Downarrow \\
\text{I.28} \\
\Downarrow \\
AB \parallel CD.
\end{array}$$

A.5 Comparison with the Lean reconstruction at XI.6

The Lean proof preserves the central metric sequence of the classical argument:

$$\text{auxiliary } DE \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

The formal expansion is concentrated in the data suppressed by the classical diagram:

- each planar construction is performed in an explicit induced plane geometry;
- cross-plane segment and angle comparisons use the spatial congruence layer;
- noncollinearity and point-in-plane obligations are explicit;
- the common plane of the two final normals is produced as a witness;
- spatial parallelism records both coplanarity and disjointness, so skew lines are excluded;
- the last planar disjointness proof uses the established neutral alternate-angle infrastructure rather than a direct call to the public I.28 wrapper.

The last point is a dependency-graph difference, not a change in the geometric target. Euclid closes the configuration by I.28; the current Lean source imports the Proposition I.27 module and invokes the lower alternate-angle criterion in the induced common plane.

A.6 Checkpoint XI.6

Through XI.6 the reconstruction remains synthetic. No coordinate, vector, scalar-product, trigonometric, or numerical-angle representation is used in the proposition proofs.

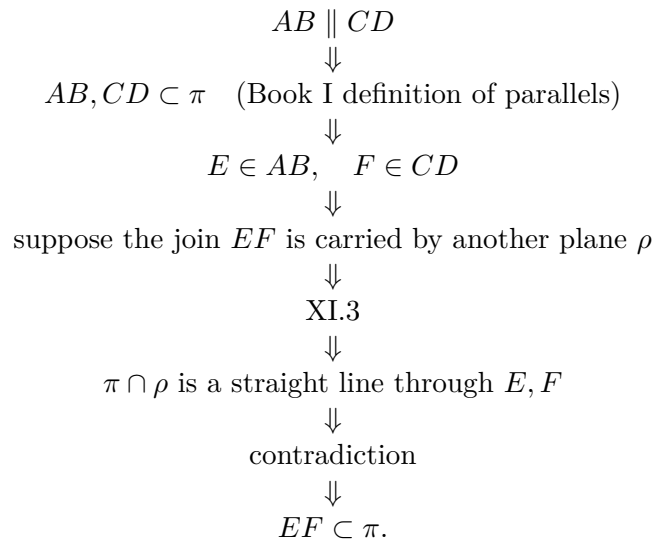
The source comparison also isolates the new structural point of XI.6: in three-space the proof of parallelism has to establish coplanarity before planar nonintersection can be invoked. Euclid obtains this by XI.5 and XI.2. The Lean theorem records the same distinction directly in the target relation `HilbertSpaceLinesParallel`, which consists of an explicit common plane together with disjointness of the two ambient carriers.

A.7 Proposition XI.7

Proposition XI.7 concerns the plane determined by two parallel straight lines. In Book I the definition of parallel lines already requires the two lines to be in one plane. Thus the reference plane is part of the hypothesis, not an additional conclusion of XI.7.

Choose a point on each of the two parallel lines and join the chosen points. Euclid argues by contradiction. He supposes that the joining straight line is carried by another, more elevated plane. A plane through that alleged join is then intersected with the reference plane. Proposition XI.3 makes the common section a straight line. Since both chosen points lie in both planes, that section passes through both of them. The contradiction is expressed in the classical text by the impossibility of two straight lines enclosing an area. The join must therefore lie in the original plane of the parallels.

The local classical DAG is



No Proposition XI.4, XI.5, or XI.6 occurs in this local proof. The explicit Book XI dependency is XI.3.

A.8 Comparison with the Lean reconstruction at XI.7

The Lean proof has a shorter incidence DAG. The formal relation `HilbertSpaceLinesParallel` already contains a witness plane σ for the two parallel carriers, together with their disjointness.

The selected points therefore lie in σ . Their distinctness follows from disjointness of the parallel carriers. After the joining carrier is constructed, `HilbertSpaceIncidence.line_in_plane` places the whole join in σ .

This final call represents the Hilbert incidence principle that two distinct points of a line lying in a plane force the entire line to lie in that plane. Consequently the formal proof does not reproduce Euclid's auxiliary second plane or his use of XI.3. This is a difference in foundational packaging, not in the geometric conclusion.

A.9 Checkpoint XI.7

Through XI.7 the reconstruction remains entirely synthetic. Proposition XI.7 uses only incidence: no order, congruence, angle theory, continuity, coordinate representation, vector formalism, or scalar product enters its proof.

The source comparison now exhibits two distinct ways to express the same closure principle. Euclid derives it by a plane-intersection contradiction; the Hilbert-style formalization uses an explicit common-plane witness for spatial parallelism and a general line-in-plane incidence axiom.

A.10 Proposition XI.8

Proposition XI.8 states that if two straight lines are parallel and one of them is perpendicular to a plane, then the other is perpendicular to the same plane.

Let $AB \parallel CD$, with AB perpendicular to the reference plane at B . Euclid lets CD meet the plane at D , joins BD , draws DE in the reference plane perpendicular to BD , makes $DE = AB$, and joins BE , AE , and AD .

The local classical dependencies are unusually informative because they cross the Euclidean parallel boundary.

- XI.7 places AB , CD , and BD in one plane.
- I.11 and I.3 supply the auxiliary perpendicular and equal segment.
- The definition of line-plane perpendicularity makes $\angle ABD$ and $\angle ABE$ right.
- I.29 converts the given parallelism into the right angle $\angle CDB$, hence $CD \perp BD$.
- I.4 and I.8 give $AD = BE$ and then show $\angle EDA$ right.
- XI.4 applied in the plane through BD, DA gives $ED \perp CD$.
- XI.4 applied again in the reference plane, to DB, DE , gives $CD \perp \pi$.

The incidence step placing CD in the plane through BD, DA uses the planarity of triangle ABD together with the common plane of AB, CD, BD . The printed diagram suppresses this plane identification.

The local classical DAG is

$$\begin{array}{c}
AB \parallel CD, \quad AB \perp \pi \\
\Downarrow \\
\text{XI.7} \\
\Downarrow \\
DE \perp BD, \quad DE = AB \\
\Downarrow \\
\text{I.29} \\
\Downarrow \\
CD \perp BD \\
\Downarrow \\
\text{I.4, I.8} \\
\Downarrow \\
ED \perp AD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
ED \perp CD \\
\Downarrow \\
\text{XI.4} \\
\Downarrow \\
CD \perp \pi.
\end{array}$$

A.11 Comparison with the Lean reconstruction at XI.8

The Lean proof preserves the synthetic metric core but reorganizes two classical steps.

First, the point D where the second parallel meets the reference plane is not assumed. It is derived from the common carrier plane of the parallels, the intersection of that plane with the reference plane, and Group IV uniqueness.

Second, the proof does not call I.29 directly. It constructs through D a line perpendicular to BD , proves by the neutral I.27 alternate-angle criterion that this line is disjoint from the first parallel, and then uses Group IV uniqueness to identify it with the second parallel.

Thus the formal factorization is

$$\text{neutral I.27} + \text{Euclidean uniqueness of parallels} \implies \text{the I.29-type conclusion needed in XI.8.}$$

The two applications of XI.4 remain visible in the production proof. The first is made in the common plane of the parallel lines; the second is made in the reference plane.

A.12 Checkpoint XI.8

Through XI.8 the reconstruction remains synthetic: no coordinates, vectors, scalar product, trigonometric functions, or numerical angle measure are introduced.

Unlike XI.4–XI.7, however, XI.8 is genuinely Euclidean in the axiomatic sense. The formal proof requires plane-local parallel uniqueness, represented by `HilbertSpaceEuclidean`. No new continuity assumption is used.

Appendix B

Hilbert: Spatial Axiomatic Architecture

B.1 Purpose and scope

This appendix records the Hilbert-side source audit for Euclid Book XI at the current checkpoint, Proposition XI.8. The purpose is architectural: identify which layers of incidence, order, congruence, and Euclidean parallelism enter at each theorem, and distinguish them from continuity assumptions.

B.2 Spatial architecture

The formal project separates the spatial hierarchy into

```
HilbertSpacePrimitive  
HilbertSpaceIncidence  
HilbertSpaceOrder  
HilbertSpaceCongruence  
HilbertSpaceEuclidean
```

and derives ordinary planar Hilbert geometry on each fixed plane via

```
PlanePoint  
PlaneLine  
PlaneGeo
```

with explicit bridges for incidence, betweenness, congruence, same rays, angles, and right angles.

This separation is essential at XI.4. There is deliberately no global `HilbertCongruence Geo` or `HilbertOrder Geo` instance installed on the ambient three-space. Planar order arguments are carried out inside a specified `PlaneGeo`; spatial congruence transports only the data needed between such plane slices.

B.3 Neutral status of XI.4 in the current hierarchy

The public theorem assumes the spatial incidence/order/congruence hierarchy but no `HilbertEuclideanPlane` instance. In particular, Hilbert Group IV – the Euclidean parallel axiom – does not occur in the proof of XI.4.

The final axiom audit reported by Lean is

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propept,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this checkpoint. XI.4 introduces no proposition-specific geometric axiom.

B.4 What XI.4 adds to the interface

The theorem forced one useful correction to the general 3D API: `HilbertLinesPerpendicularAt` now carries an explicit noncollinearity witness. Consequently distinctness of perpendicular carriers is derived by `hilbert_linesPerpendicularAt_ne` rather than supplied as a separate hypothesis.

The universal target is represented by

```
HilbertLinePerpendicularPlaneAt
```

which quantifies over every incidence line of the plane through the foot. This is a direct synthetic representation and introduces no metric structure.

B.5 Checkpoint XI.4

At XI.4 Hilbert's role is therefore not a one-to-one proposition-number match. It is the axiomatic architecture in which Euclid's spatial proof can be made explicit. The most significant result of the audit is the boundary: XI.4 is completed in the current incidence/order/congruence 3D hierarchy without Group IV and without a new continuity assumption.

B.6 Incidence axiom I.6 and Proposition XI.7

Proposition XI.7 isolates a much smaller part of the spatial hierarchy than XI.4–XI.6. The decisive Hilbert incidence principle is represented in the current formal interface by

```
HilbertSpaceIncidence.line_in_plane
```

Its geometric content is: if two distinct points of a line lie in a plane, then the whole line lies in that plane. In the numbering adopted by the current project this is Hilbert incidence axiom I.6.

The public XI.7 theorem assumes only

```
[HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[HilbertSpacePrimitive Geo]
[HilbertSpaceIncidence Geo]
```

It does not assume `HilbertSpaceOrder`, `HilbertSpaceCongruence`, or an induced `PlaneGeo` instance. The Euclidean parallel axiom, Group IV, and continuity are therefore absent from the proof architecture.

B.7 Spatial parallelism as reusable incidence data

The relation

```
HilbertSpaceLinesParallel
```

is now part of `Hilbert3DInterface`. It packages two assertions: a common carrier plane and disjointness of the two ambient lines. The common plane is essential because ambient disjointness alone would also admit skew lines.

This notion was introduced during the XI.6 development and promoted to the general 3D interface before XI.7. Proposition XI.7 is the first subsequent Book XI theorem that consumes the common-plane witness directly.

B.8 Axiomatic status of XI.7

The final audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This result concerns global Lean axiom constants. It does not remove the explicit Hilbert incidence structures from the theorem signature: those structures remain hypotheses of the theorem. What the audit establishes is that no additional globally declared axiom constant occurs in the proof term.

B.9 Checkpoint XI.7

At XI.7 the spatial reconstruction reaches a pure incidence theorem. The formal proof needs neither Group IV nor continuity, and it introduces no new proposition-specific geometric or spatial axiom. The decisive spatial content is exactly the combination of a common-plane witness for parallel lines and Hilbert’s line-in-plane incidence principle.

B.10 Group IV becomes active at Proposition XI.8

Proposition XI.8 is the first theorem in the present Book XI sequence whose production signature assumes

```
[HSE : HilbertSpaceEuclidean Geo]
```

The class is the spatial representation of Hilbert Group IV. Its parallel-uniqueness field is explicitly restricted to one ambient plane: through a point of that plane outside a given line, two lines in the same plane that are disjoint from the given line must coincide.

This formulation is essential in three-space. The project does not install a global

```
HilbertEuclideanPlane Geo
```

instance on the ambient spatial geometry. Instead, every fixed plane inherits the planar Euclidean interface through

```
planeGeo_linesDisjoint_iff_ambient
planeGeoHilbertEuclidean
```

The first theorem identifies planar disjointness in a fixed `PlaneGeo` with ambient disjointness of the underlying spatial lines. The second packages the plane-local Group IV field as the existing planar `HilbertEuclideanPlane` instance.

B.11 Neutral I.27 and Euclidean uniqueness in XI.8

The formal replacement for Euclid’s I.29 step exposes the logical boundary between neutral and Euclidean geometry.

Inside the common plane σ of the parallel lines, a line p is constructed through D perpendicular to the transversal $d = BD$. The established I.27 alternate-angle theorem shows that p is disjoint from the first parallel l . This part uses order and congruence but not Group IV.

The given second parallel m is also disjoint from l . Both p and m pass through D and lie in σ . Only now is Group IV used: uniqueness forces $p = m$. Therefore $m \perp d$.

The formal proof consequently separates

existence/nonintersection from Euclidean uniqueness.

B.12 Axiomatic status of XI.8

The final audit command is

```
import CGJteamLab.Proposition11_8

#print axioms Geometry.euclid_proposition_11_8
```

and Lean reports

```
'Geometry.euclid_proposition_11_8' depends on axioms: [propext,
Classical.choice,
Geometry.bookZero_nullSegment1,
Quot.sound]
```

These are inherited global Lean/project dependencies. No new proposition-specific global axiom constant is introduced.

This audit output must be distinguished from the theorem’s explicit geometric hypotheses. The typeclass `HilbertSpaceEuclidean Geo` is a genuine mathematical assumption of XI.8: it represents Group IV. It is passed as a theorem parameter, rather than appearing as an additional global Lean axiom constant.

No continuity assumption is used.

B.13 Checkpoint XI.8

At XI.8 the spatial hierarchy first crosses from the neutral incidence/order/congruence layers into the Euclidean parallel layer. The transition is plane-local: Group IV is applied only inside a specified ambient plane and is inherited by its induced `PlaneGeo` geometry.

The proof remains synthetic. Its Euclidean content is exactly the uniqueness step required to turn neutral nonintersection into the I.29-type perpendicularity conclusion.

Appendix C

Borsuk–Szmielew: Axiomatic Layers and Spatial Geometry

C.1 Scope

This appendix records the Book XI audit against Karol Borsuk and Wanda Szmielew, *Foundations of Geometry*, through Proposition XI.8. Their primitive language includes points, lines, planes, betweenness, and equidistance. Their axioms are organized into incidence, order, congruence, continuity, and the Euclidean axiom.

C.2 Congruence results relevant to XI.4

The Book I audit already identifies the triangle-congruence results used in the XI.4 argument within the pre-Euclidean congruence layer. In particular:

- the SSS theorem corresponding to Euclid I.8 occurs as Borsuk–Szmielew Theorem 21;
- right-angle and perpendicular constructions belong to their synthetic congruence development and are not formulated by numerical angle measure.

The formal XI.4 proof likewise treats SAS/SSS and right-angle transport as congruence infrastructure rather than as consequences of the Euclidean parallel axiom.

C.3 Spatial geometry

Borsuk–Szmielew develop three-dimensional geometry with points, lines, planes, perpendicular lines, perpendicular projection, and later an absolute coordinate system in space. In that spatial development a line perpendicular to a plane is part of the geometric language before coordinates are introduced as a later representation.

At the current checkpoint no theorem number has been identified in the audited source whose statement is exactly Euclid XI.4. Accordingly the Lean theorem is not attributed here to a specific Borsuk–Szmielew theorem.

C.4 Axiomatic comparison

For XI.4 the comparison is:

$$\begin{array}{c}
 \text{incidence and order in planes} \\
 + \\
 \text{synthetic segment and angle congruence} \\
 + \\
 \text{spatial incidence/congruence transport} \\
 \Downarrow \\
 \text{line–plane perpendicularity.}
 \end{array}$$

The formal proof does not use the Euclidean parallel axiom. This agrees with the placement of the triangle-congruence ingredients below the Euclidean layer in the Borsuk–Szmielw organization.

C.5 Checkpoint XI.4

The current audit records no exact Borsuk–Szmielw counterpart for XI.4 itself. Exact theorem-number correspondence should be added only if it is located in the source.

C.6 Proposition XI.7 and incidence Axiom I7

For XI.7 the Borsuk–Szmielw source contains an exact structural counterpart of the decisive Hilbert incidence clause.

Their Axiom I7 states that for a line L and a plane P , if there exist two distinct points a, b belonging both to L and to P , then

$$L \subset P.$$

This is exactly the line-in-plane principle used in the formal proof of XI.7. Borsuk–Szmielw also classify Axioms I1–I4 as plane axioms and Axioms I5–I9 as space axioms. Hence I7 belongs to the spatial incidence layer.

Their general organization is especially transparent here: the first group consists of incidence axioms concerning points, lines, and planes; order, congruence, continuity, and the Euclidean axiom are introduced only in later groups. Proposition XI.7 therefore lies strictly below all of those stronger layers.

The correspondence with the Lean proof is

$$\begin{array}{c}
 E, F \in \sigma, \quad E \neq F \\
 \Downarrow \\
 \text{line } n \text{ through } E, F \\
 \Downarrow \\
 \text{Borsuk–Szmielw I7 / Hilbert I.6} \\
 \Downarrow \\
 n \subset \sigma.
 \end{array}$$

No Borsuk–Szmielw theorem number is needed here: the relevant source item is the incidence axiom itself.

C.7 Checkpoint XI.7

At XI.7 the comparison becomes sharper than at XI.4. The source contains an explicit spatial-incidence axiom whose content matches the decisive formal step exactly.

Thus XI.7 requires only incidence structure in the Borsuk–Szmielew hierarchy. It does not require their axioms of order, congruence, continuity, or the Euclidean axiom. This agrees with the Lean theorem, which assumes only the ambient incidence classes and whose axiom audit reports no additional global axiom constants.

C.8 Proposition XI.8 and the Euclidean parallel boundary

No exact Borsuk–Szmielew theorem number matching the full statement of Euclid XI.8 has been identified in the audited source. The source does, however, give a precise axiomatic placement for the parallel-theoretic step on which XI.8 depends.

The Book I source audit places the I.27-type implication

$$\text{appropriate angle equality} \implies \text{nonintersection}$$

in the congruence development before continuity and before the Euclidean axiom.

After the Euclidean axiom E is introduced, the source obtains uniqueness of the parallel and develops the converse Euclidean parallel-angle theory. In particular, the material corresponding to the I.29 direction lies on the Euclidean side of this boundary.

This is exactly the split exposed by the Lean proof of XI.8. The proposition-local helper first uses the neutral alternate-angle theorem to construct a nonintersecting line. The spatial Group IV class then supplies uniqueness in the fixed plane and identifies that line with the given parallel.

Thus the source-layer comparison is

$$\begin{array}{c} \text{incidence} + \text{order} + \text{congruence} \\ \Downarrow \\ \text{I.27-type nonintersection} \\ \Downarrow \\ + \text{Euclidean axiom } E / \text{Group IV} \\ \Downarrow \\ \text{unique parallel and I.29-type converse} \\ \Downarrow \\ \text{XI.8 perpendicularity transport.} \end{array}$$

The metric SAS/SSS and right-angle ingredients used around this step remain in the pre-Euclidean congruence layer. The new axiomatic input is parallel uniqueness, not a new metric or continuity principle.

C.9 Checkpoint XI.8

At XI.8 the Borsuk–Szmielew comparison records the first Book XI checkpoint in this reconstruction that requires the Euclidean axiom layer. No new continuity assumption is needed.

The source does not justify assigning a one-to-one Borsuk–Szmielew theorem number to XI.8 itself. The durable source observation is instead the same boundary made explicit by the formal API: neutral nonintersection precedes E , while the converse parallel-angle information requires Euclidean uniqueness.

Bibliography

- [1] Euclid, *The Thirteen Books of Euclid's Elements*, edited and translated by Sir Thomas L. Heath, 3 vols., Dover Publications, New York, 1956. (Reprint of the second edition, Cambridge University Press, 1926.)
- [2] David Hilbert, *Grundlagen der Geometrie*, B. G. Teubner, Leipzig, 1899.
- [3] Karol Borsuk and Wanda Szmielew, *Foundations of Geometry: Euclidean and Bolyai-Lobachevskian Geometry, Projective Geometry*, North-Holland Publishing Company, Amsterdam, 1960.
- [4] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.